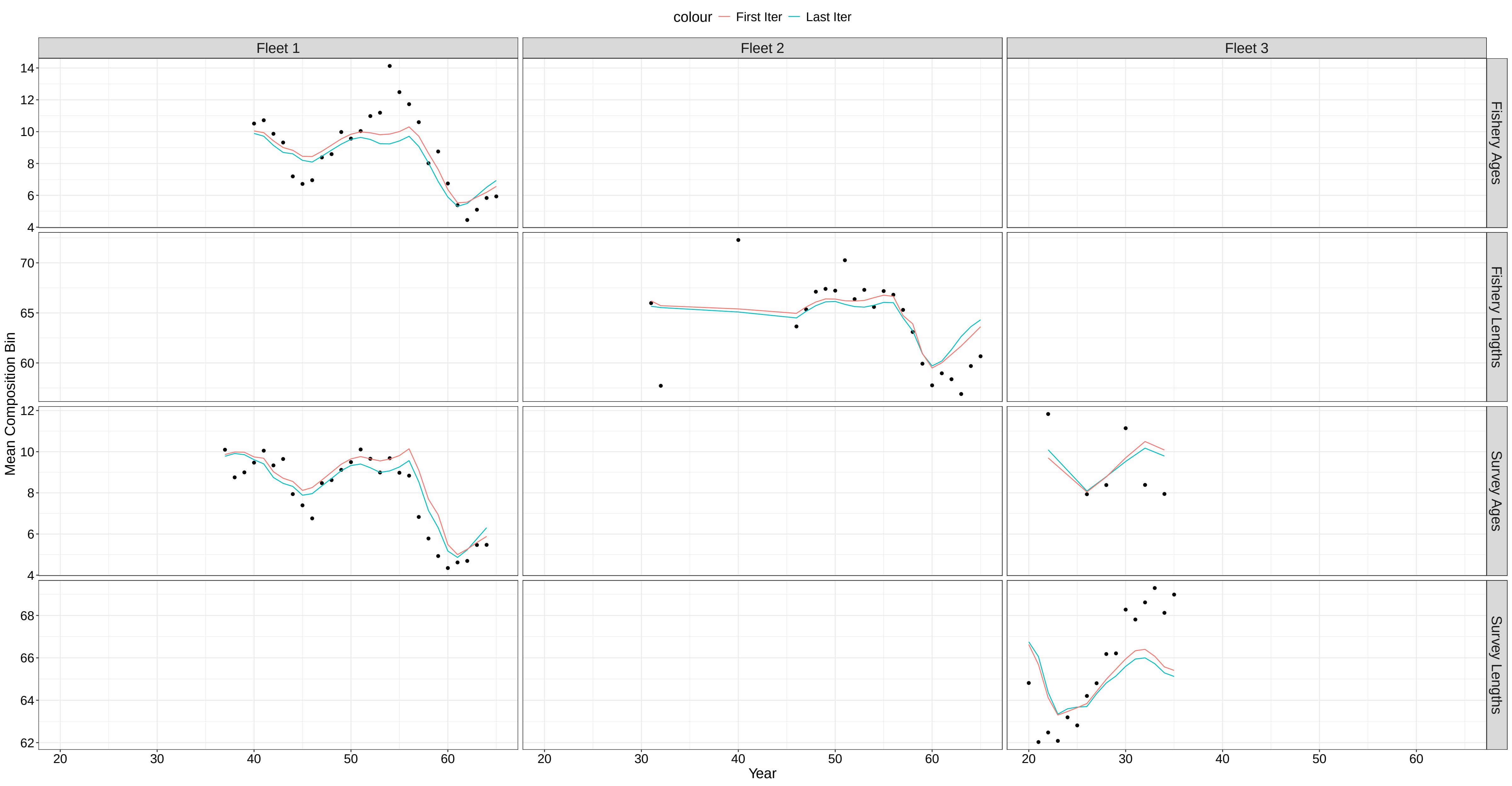
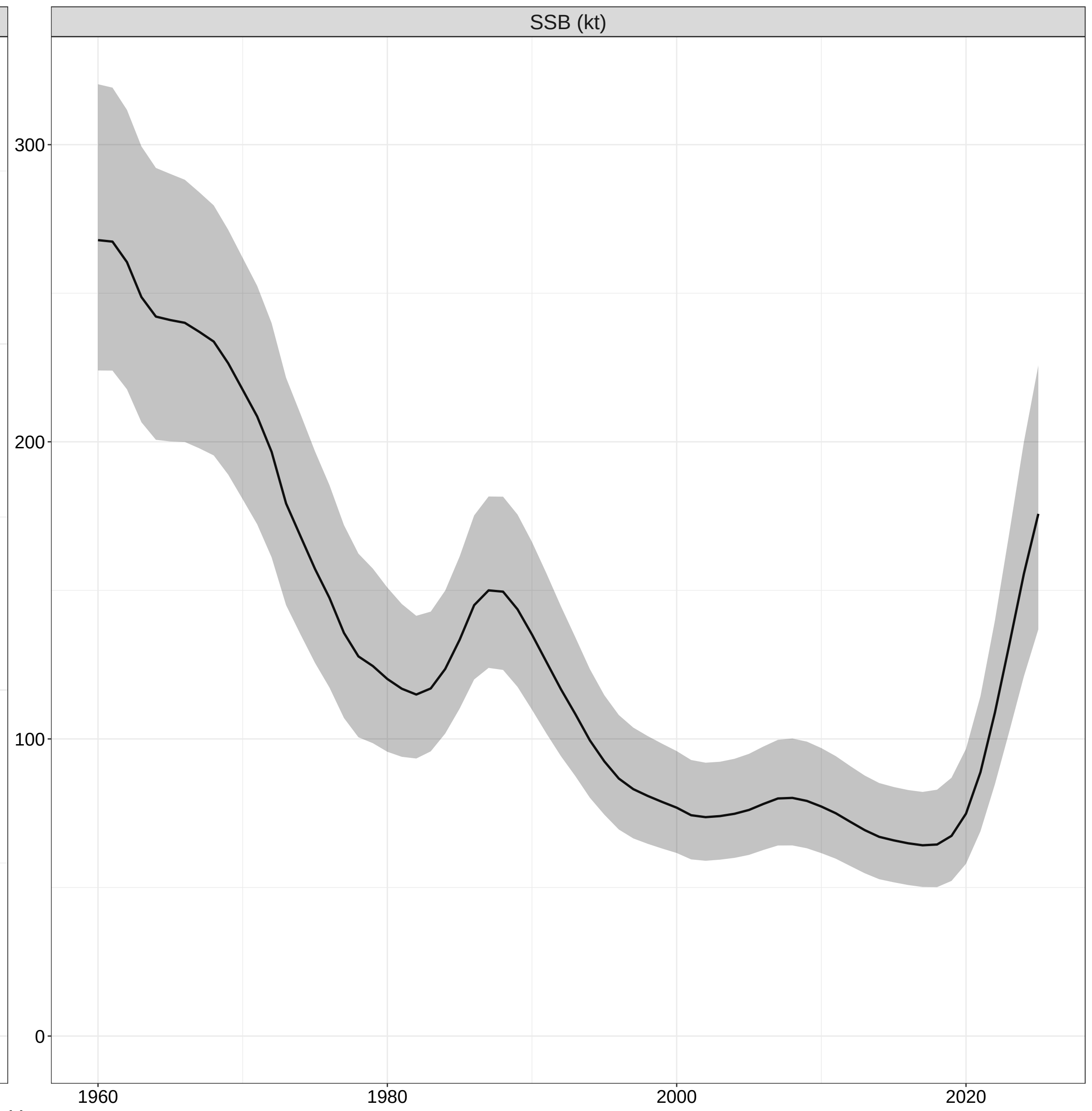
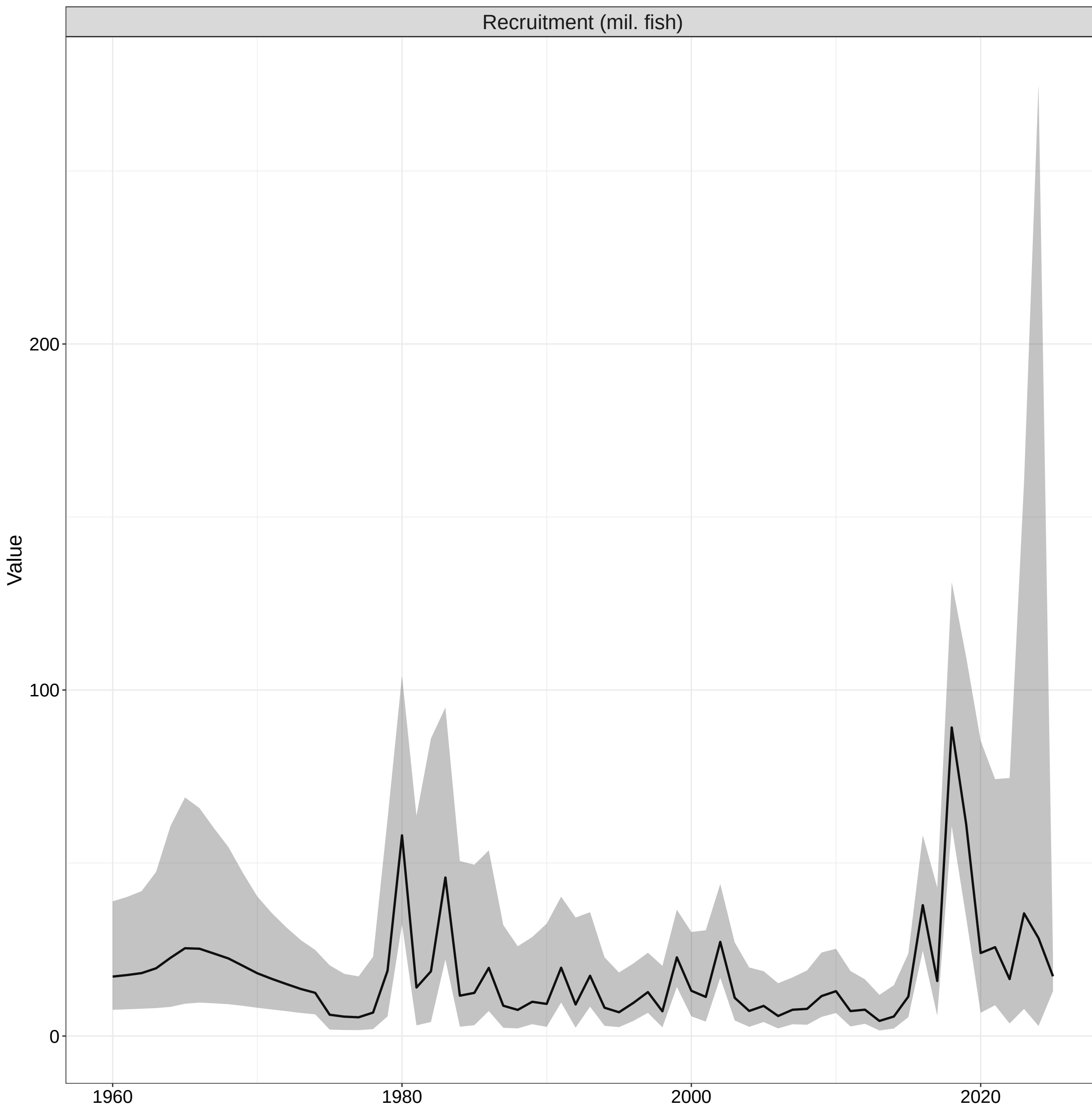
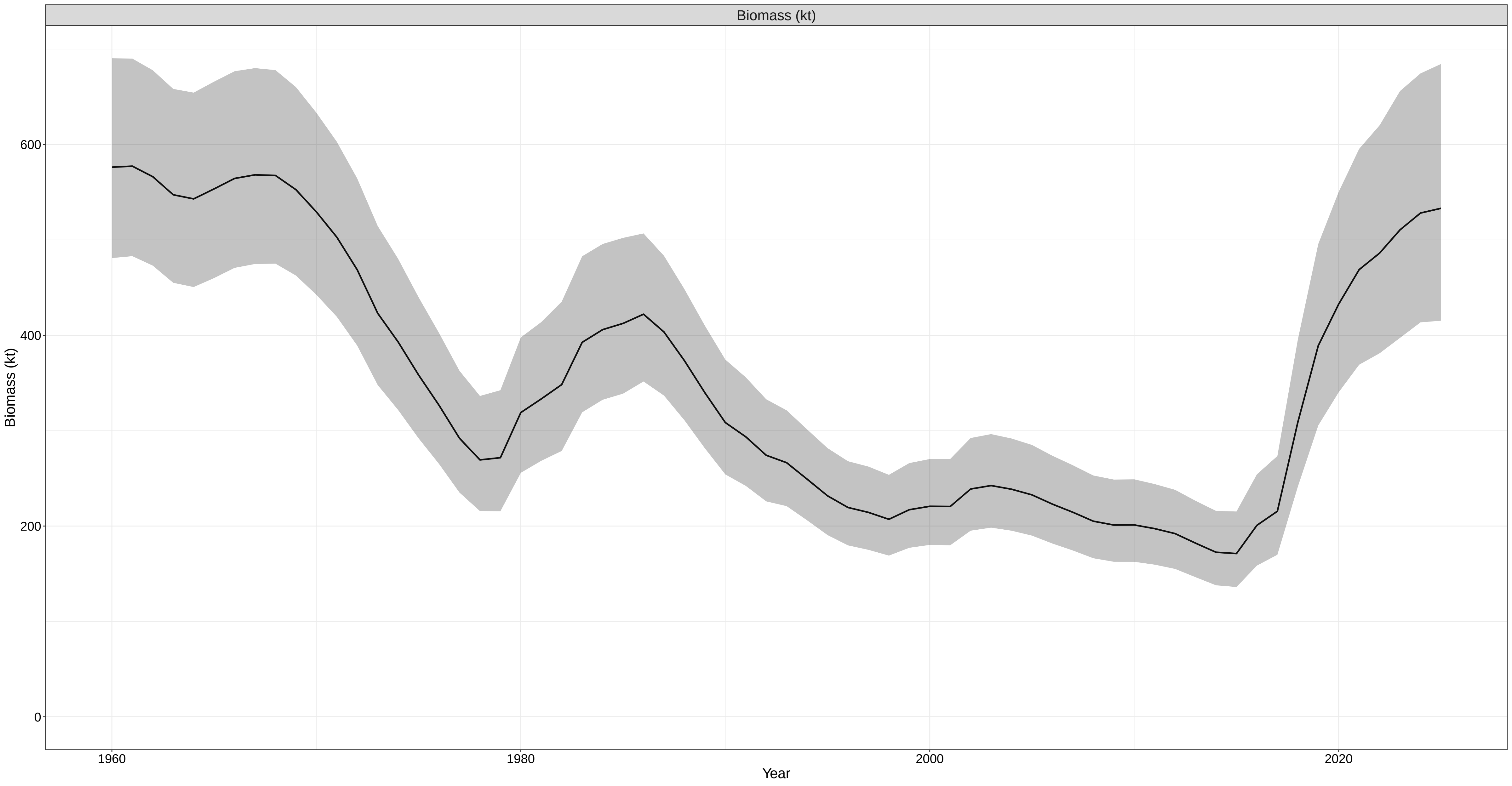


	Comp_Type	Init_Weight	Final_Weight
1	LLF_Age_F	0.8261073	0.7451677
2	LLF_Age_M	NA	NA
3	LLS_Age_F	3.7922454	3.3234193
4	LLS_Age_M	NA	NA
5	JPN_LLS_Age_F	1.3168111	0.9442567
6	JPN_LLS_Age_M	NA	NA
7	TF_Len_F	0.3164859	0.3636786
8	TF_Len_M	0.2293966	NA
9	JPN_LLS_Len_F	1.2777281	0.7764835
10	JPN_LLS_Len_M	0.8575195	NA



	Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
1	1	1	175.75182	336.29499	0.0556959785086657	33.33876	128.04473	0.07677	1.37258	0.52261	0.72545

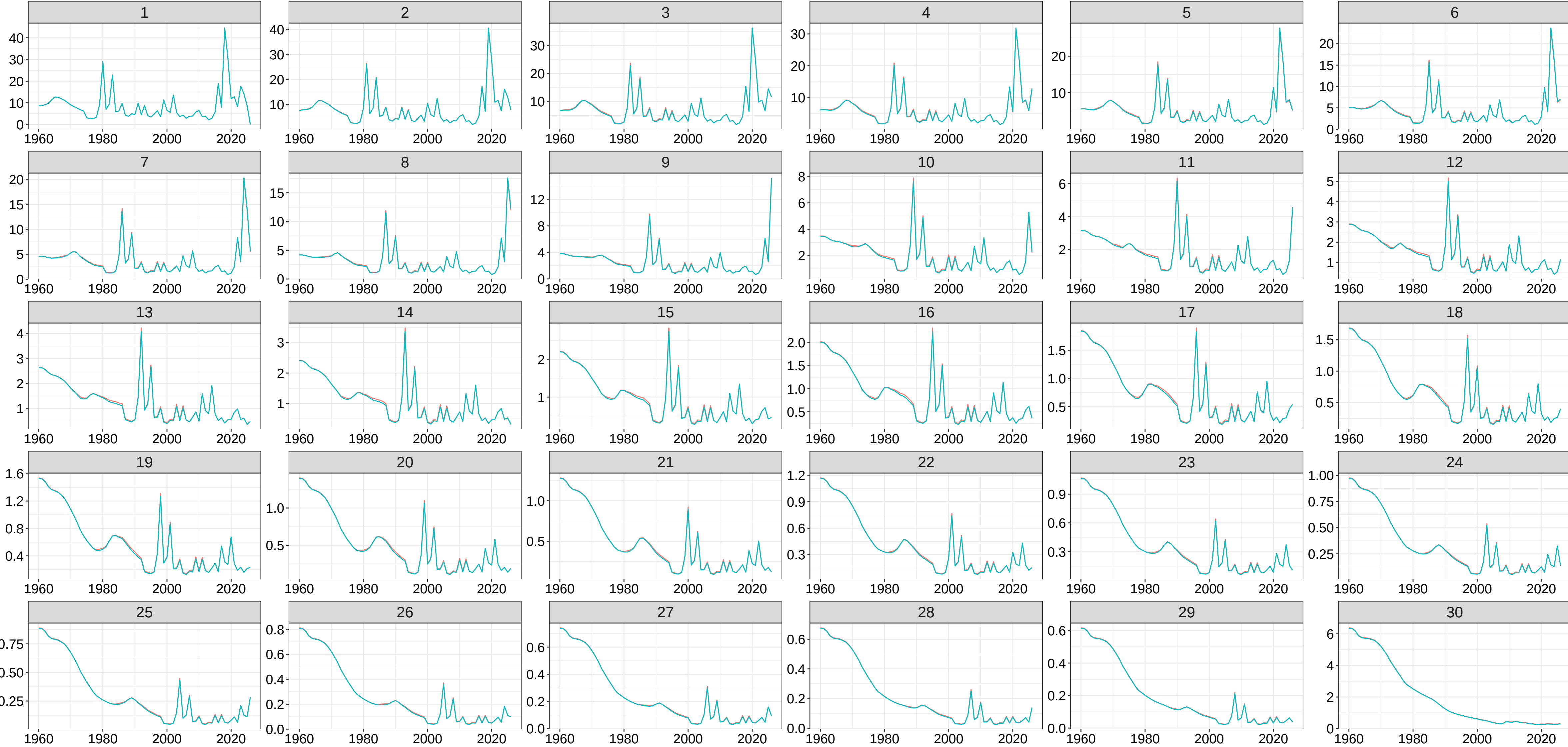




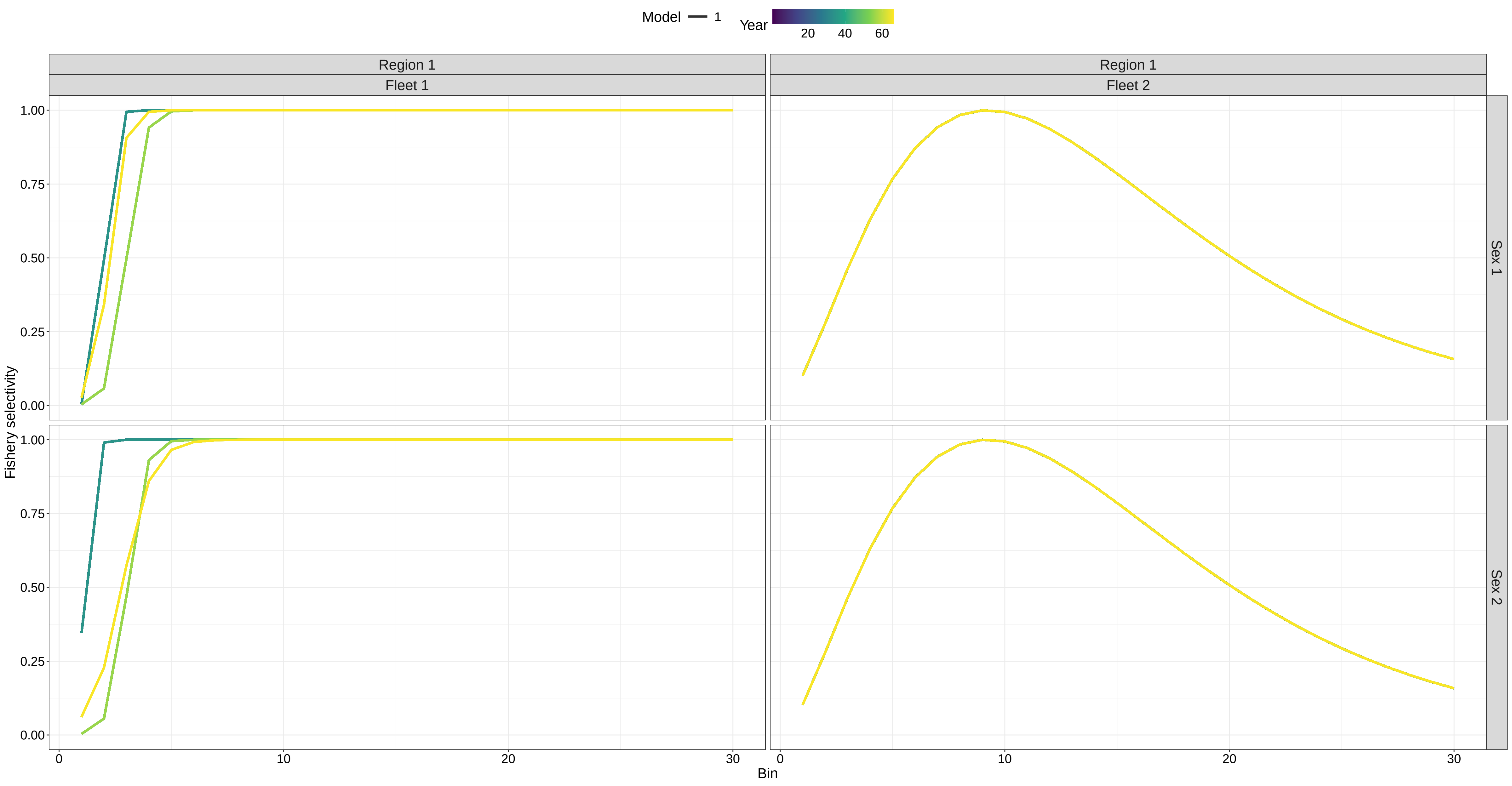
Numbers-at-Age

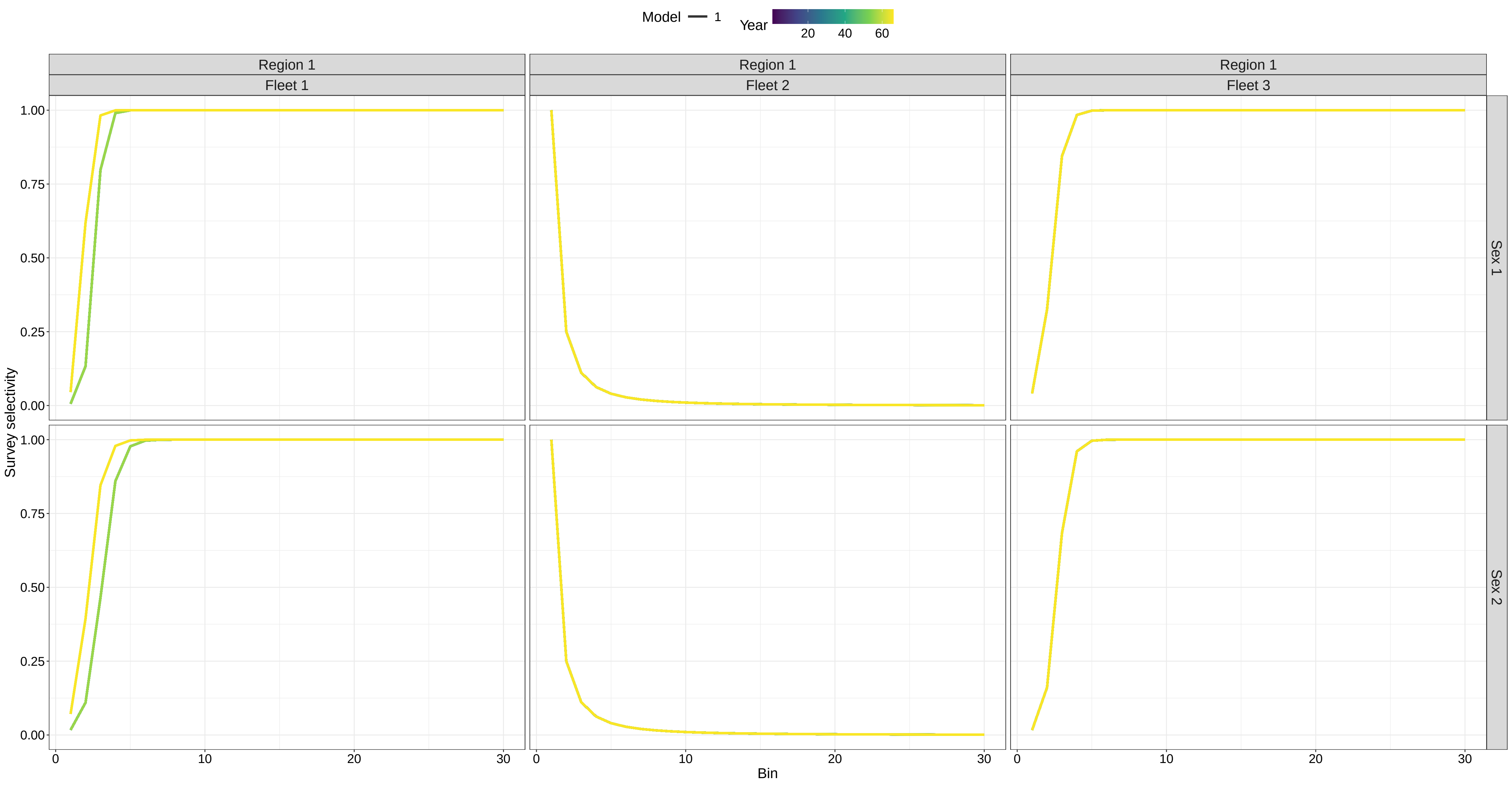
Sex — Female — Male

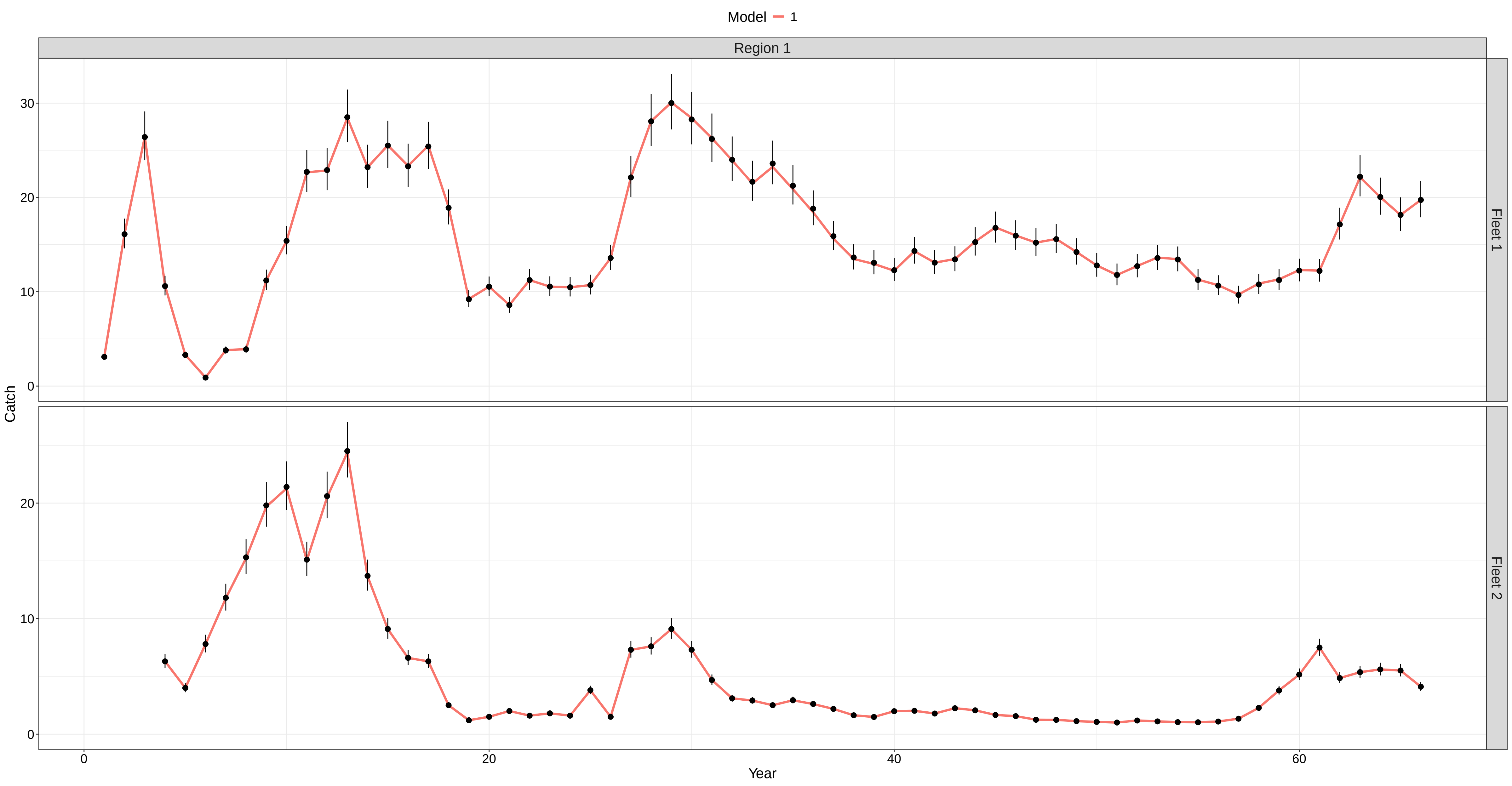
Numbers-at-age (millions)



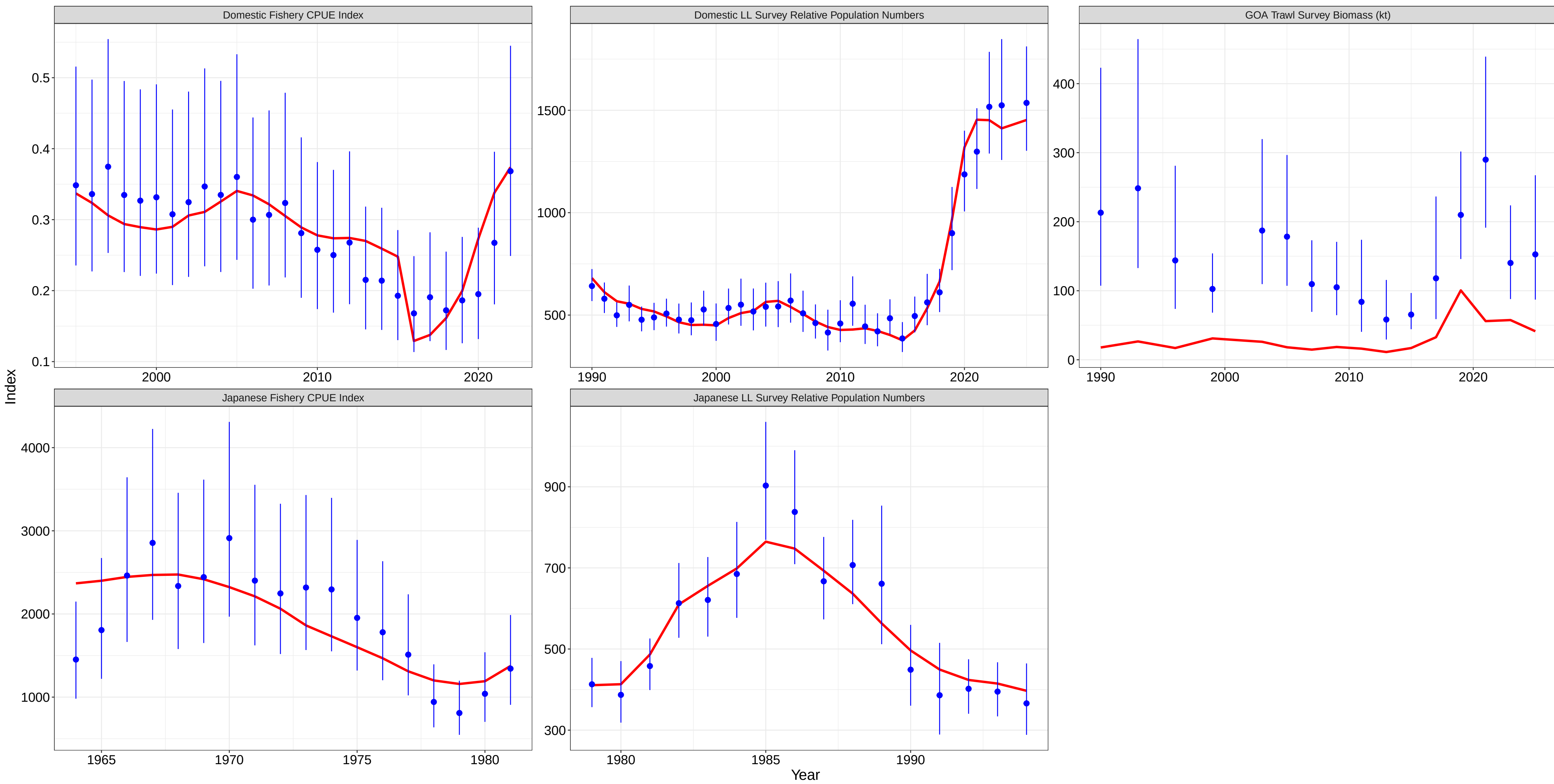
Year



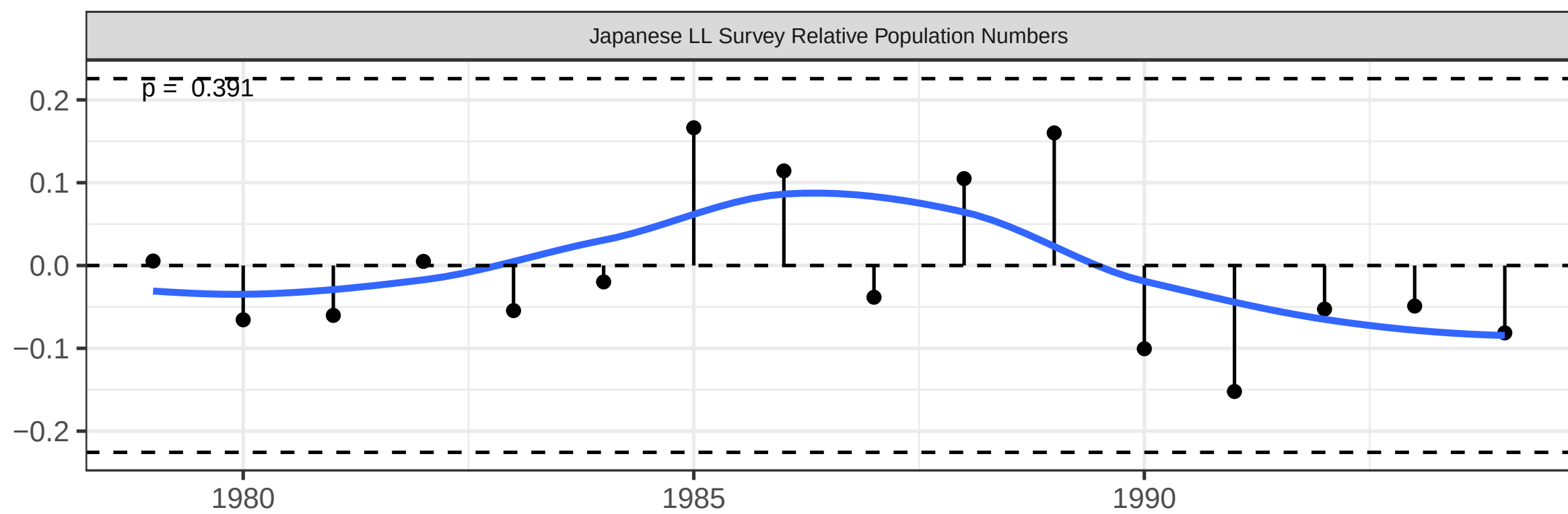
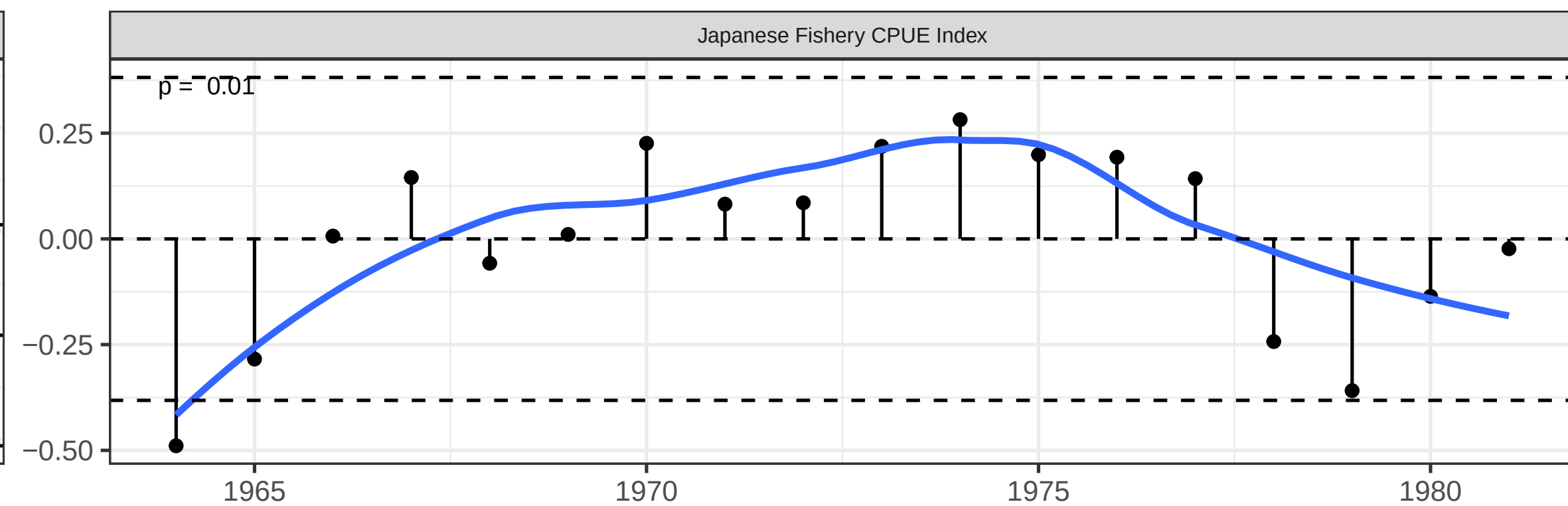
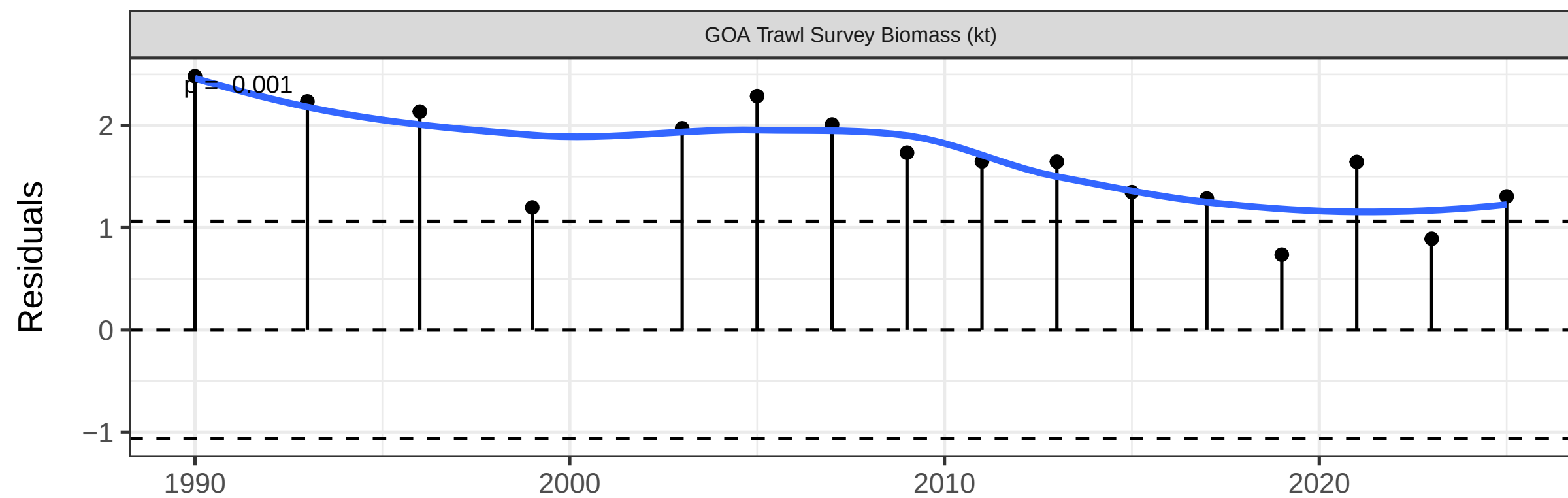
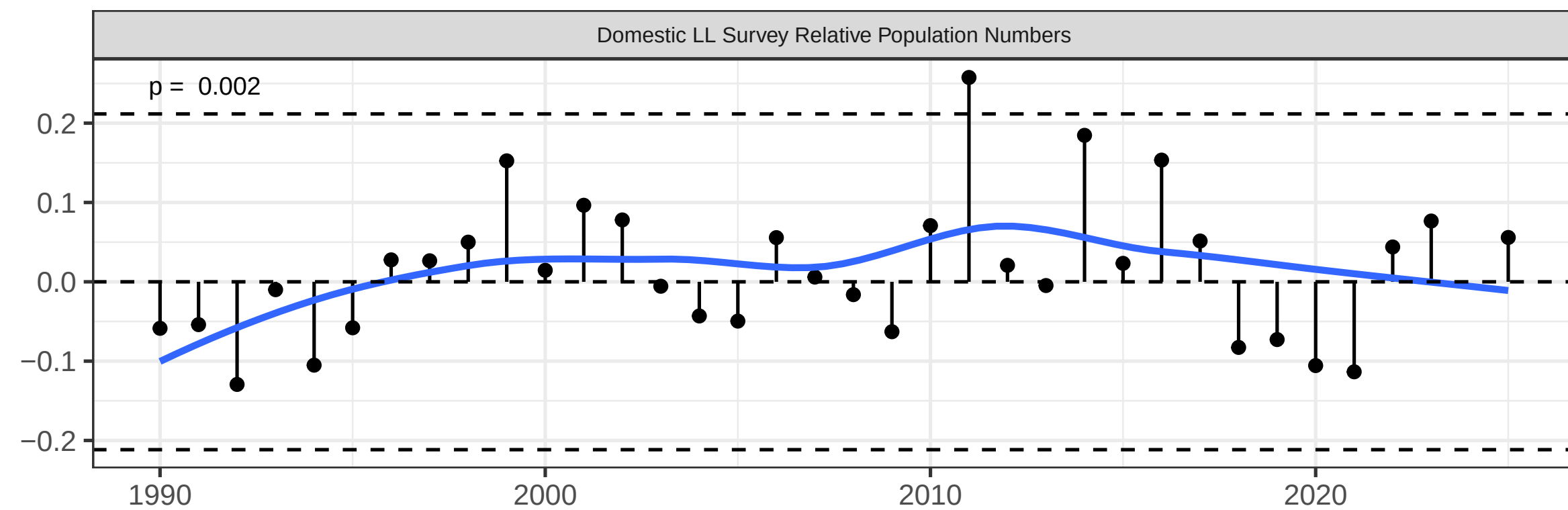
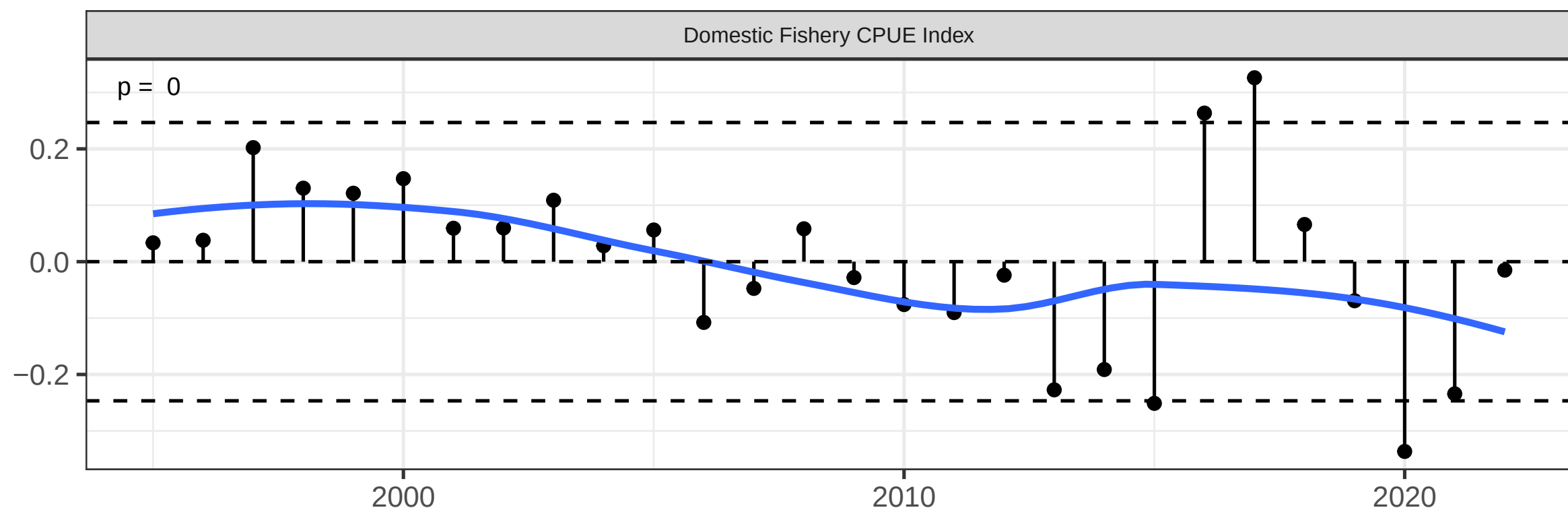




Index Fits

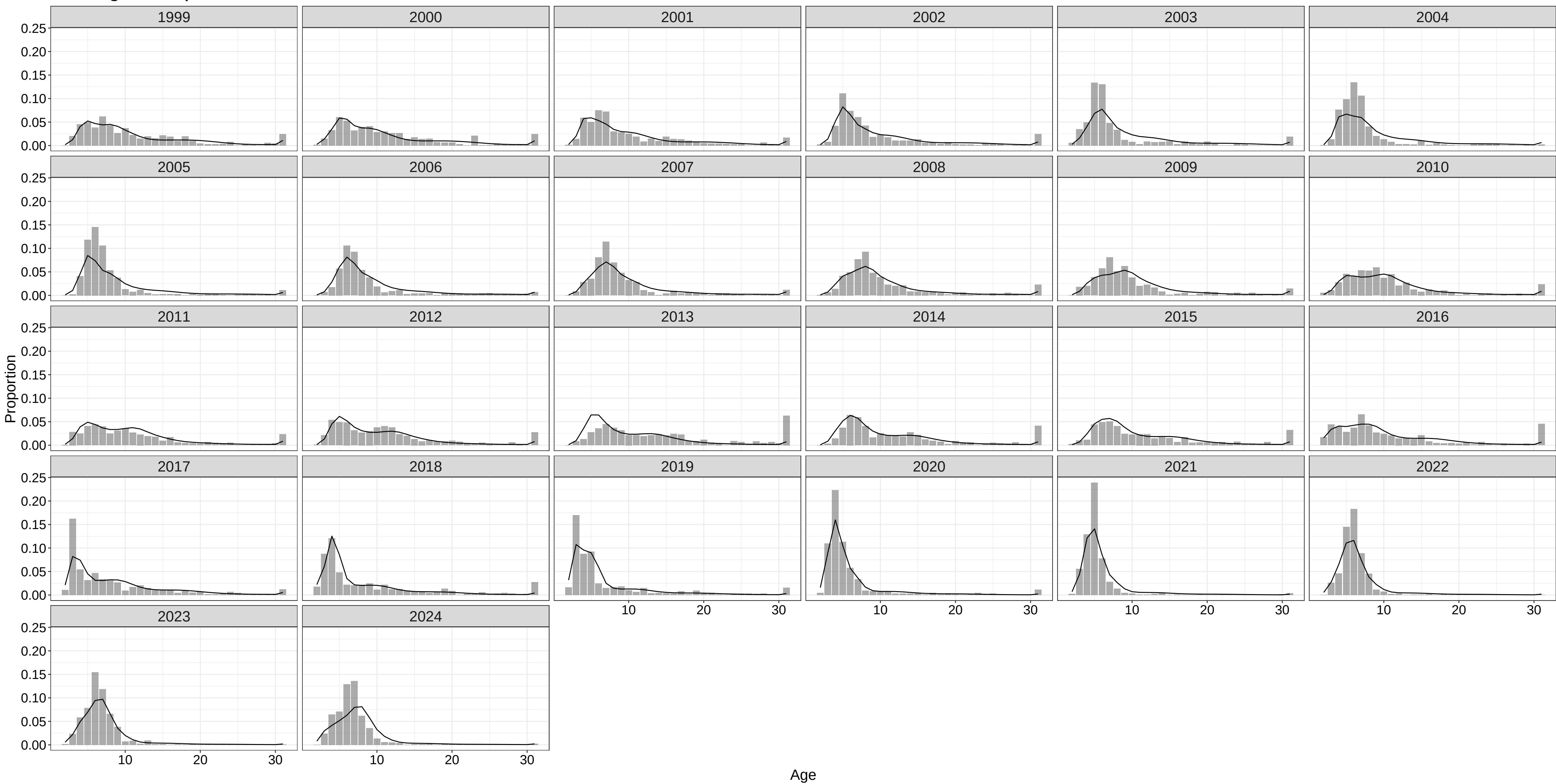


Index Residuals Runs Test

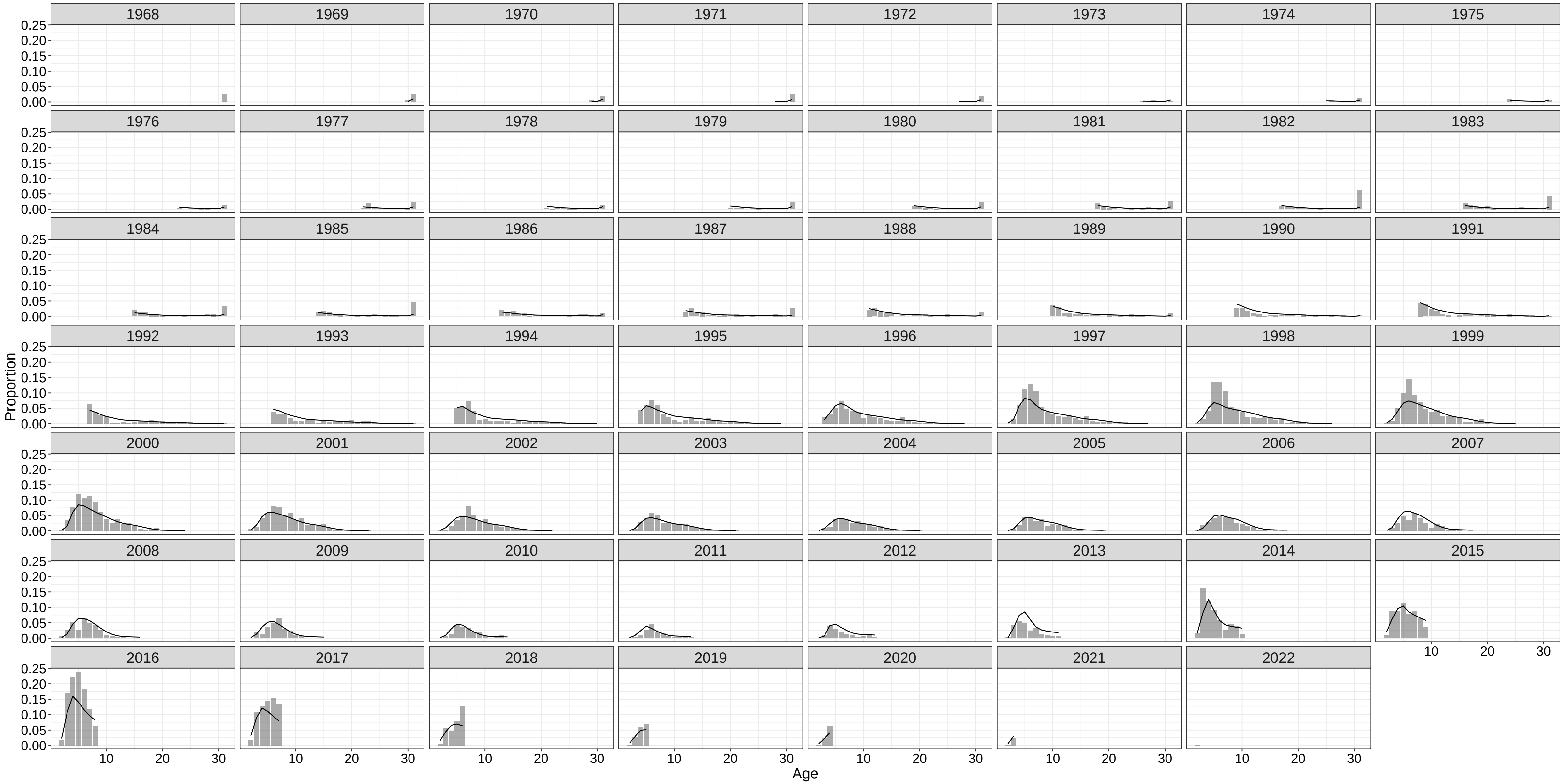


Year

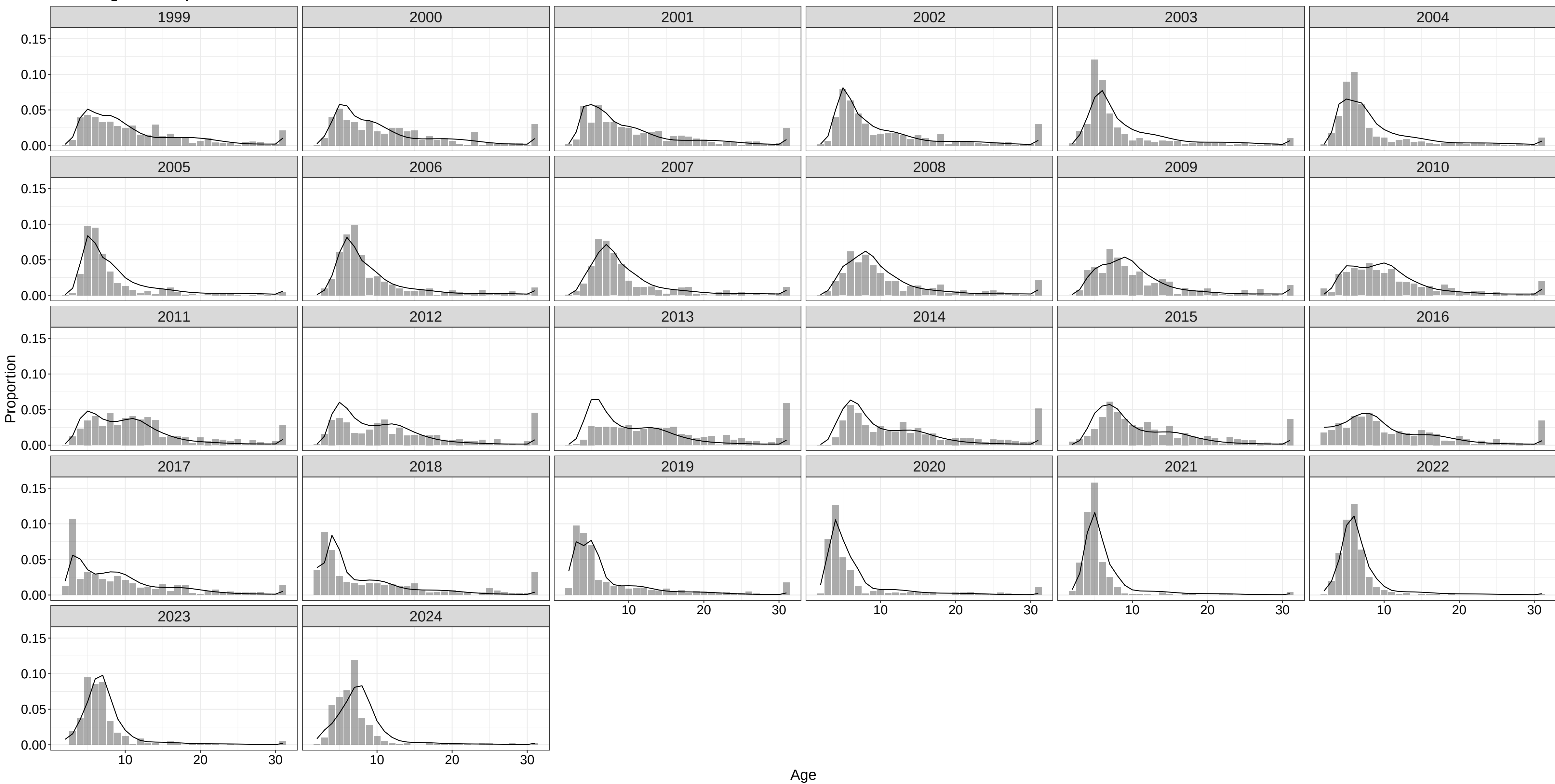
LLF Age Comps Female



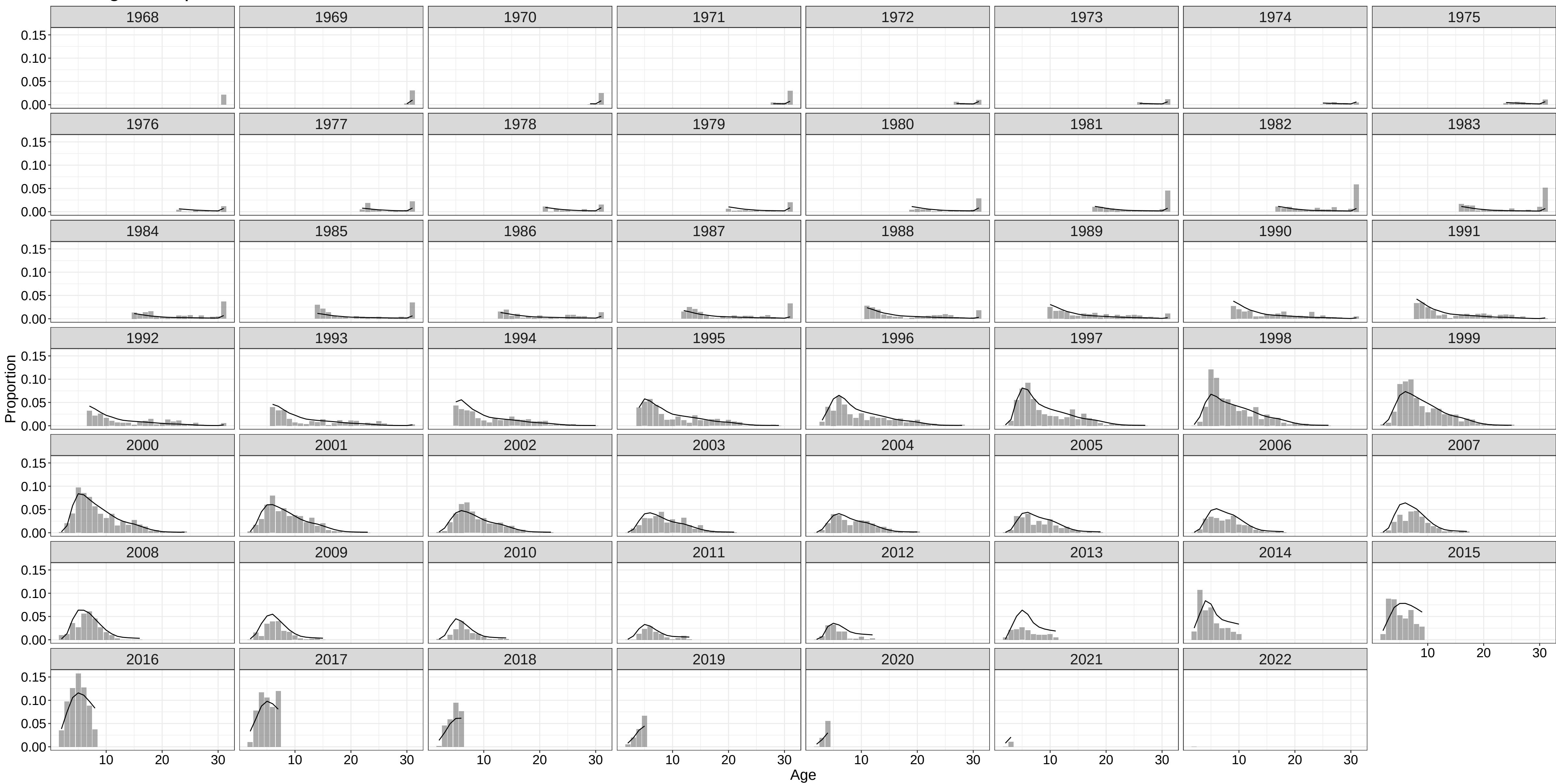
LLF Age Comps Cohort Fits Female



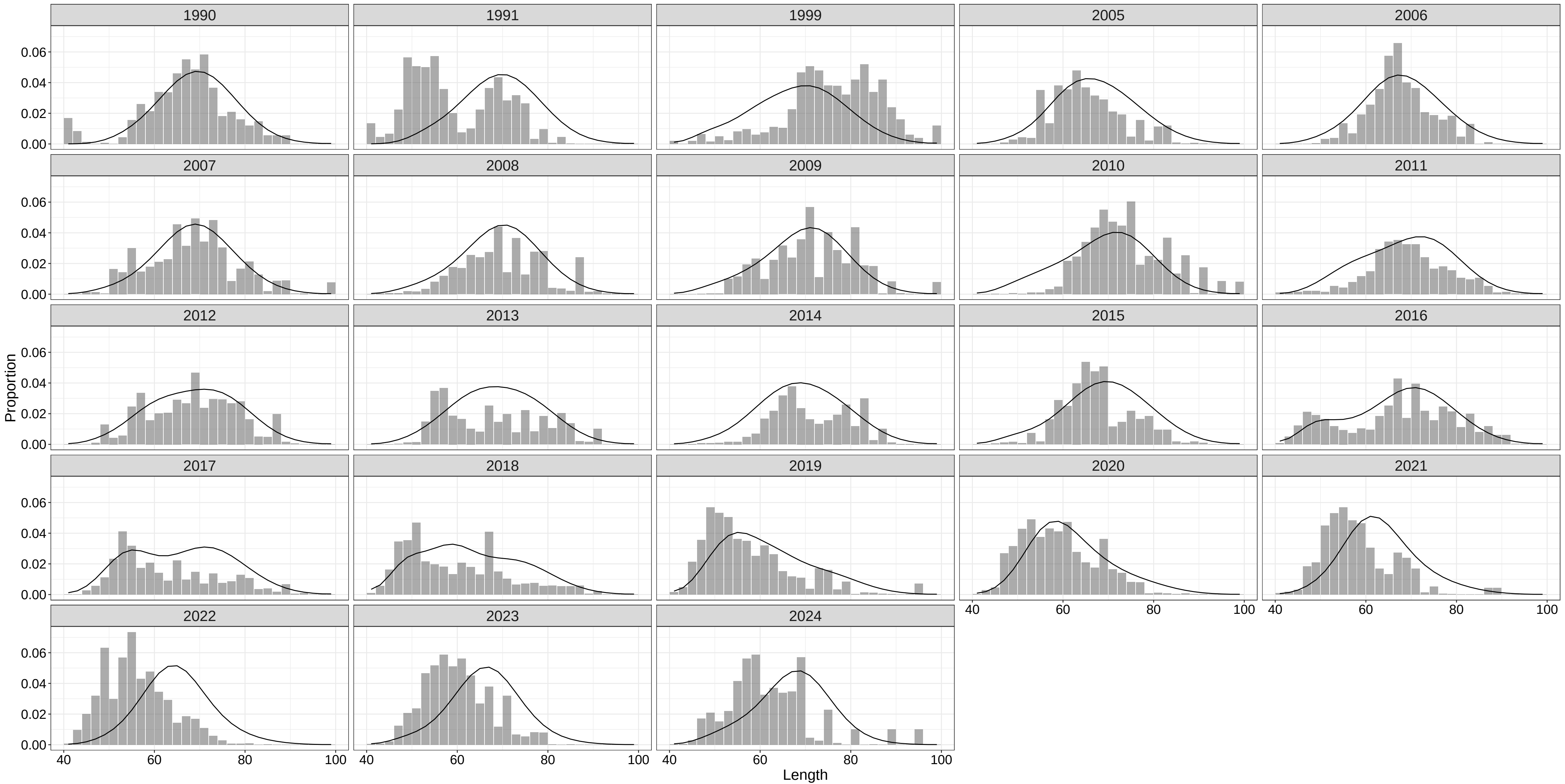
LLF Age Comps Male



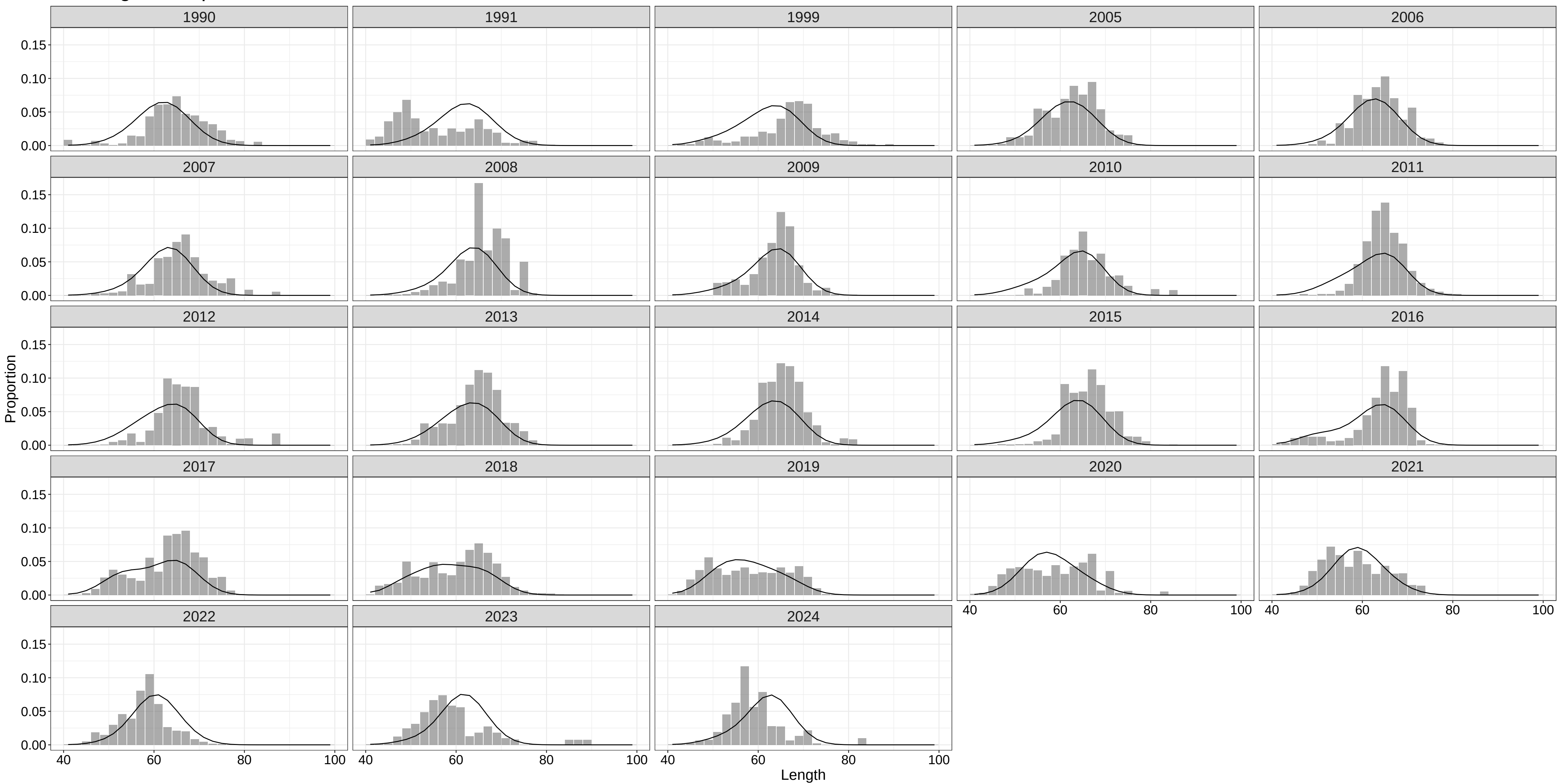
LLF Age Comps Cohort Fits Male



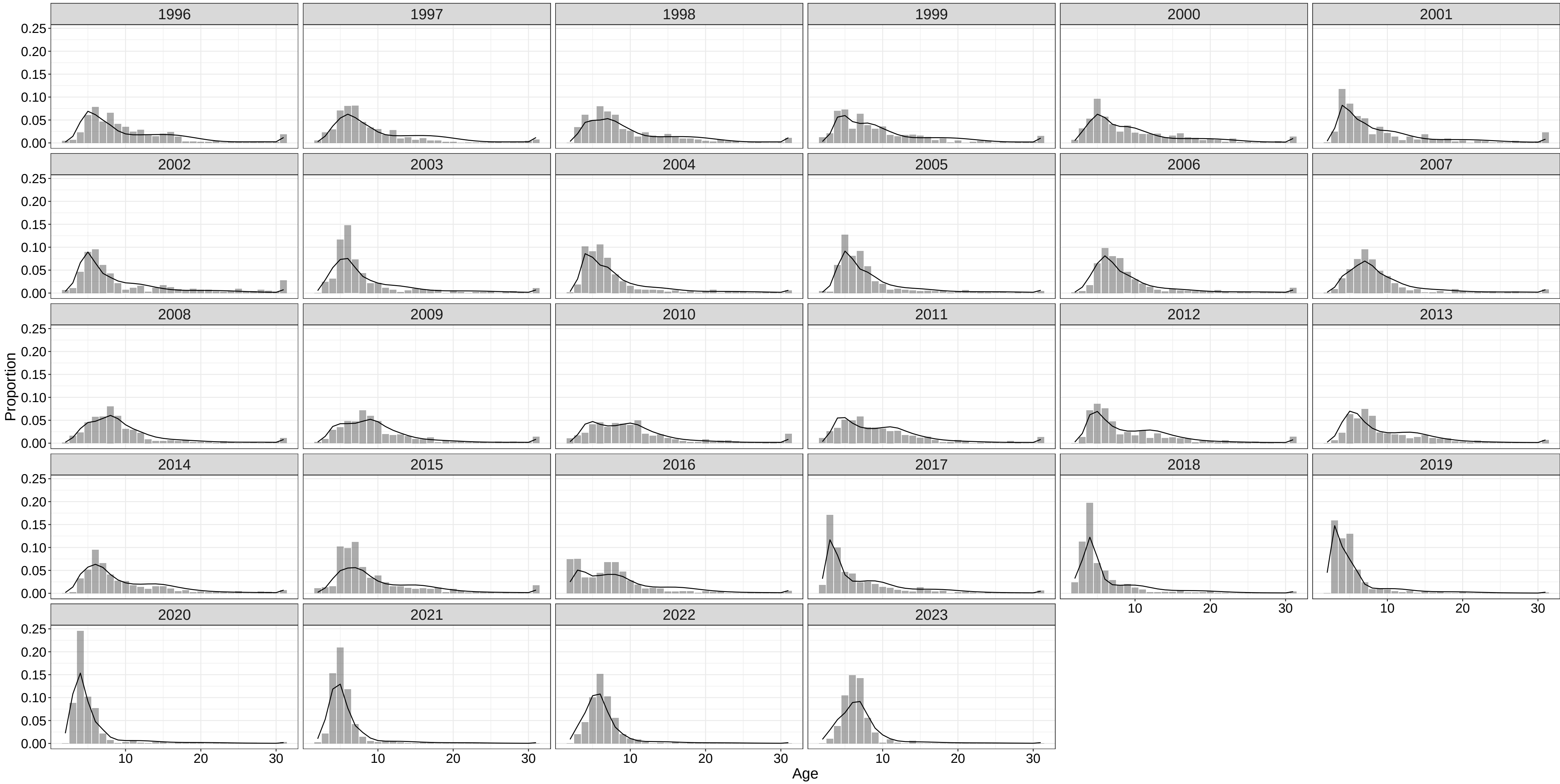
TF Length Comps Female



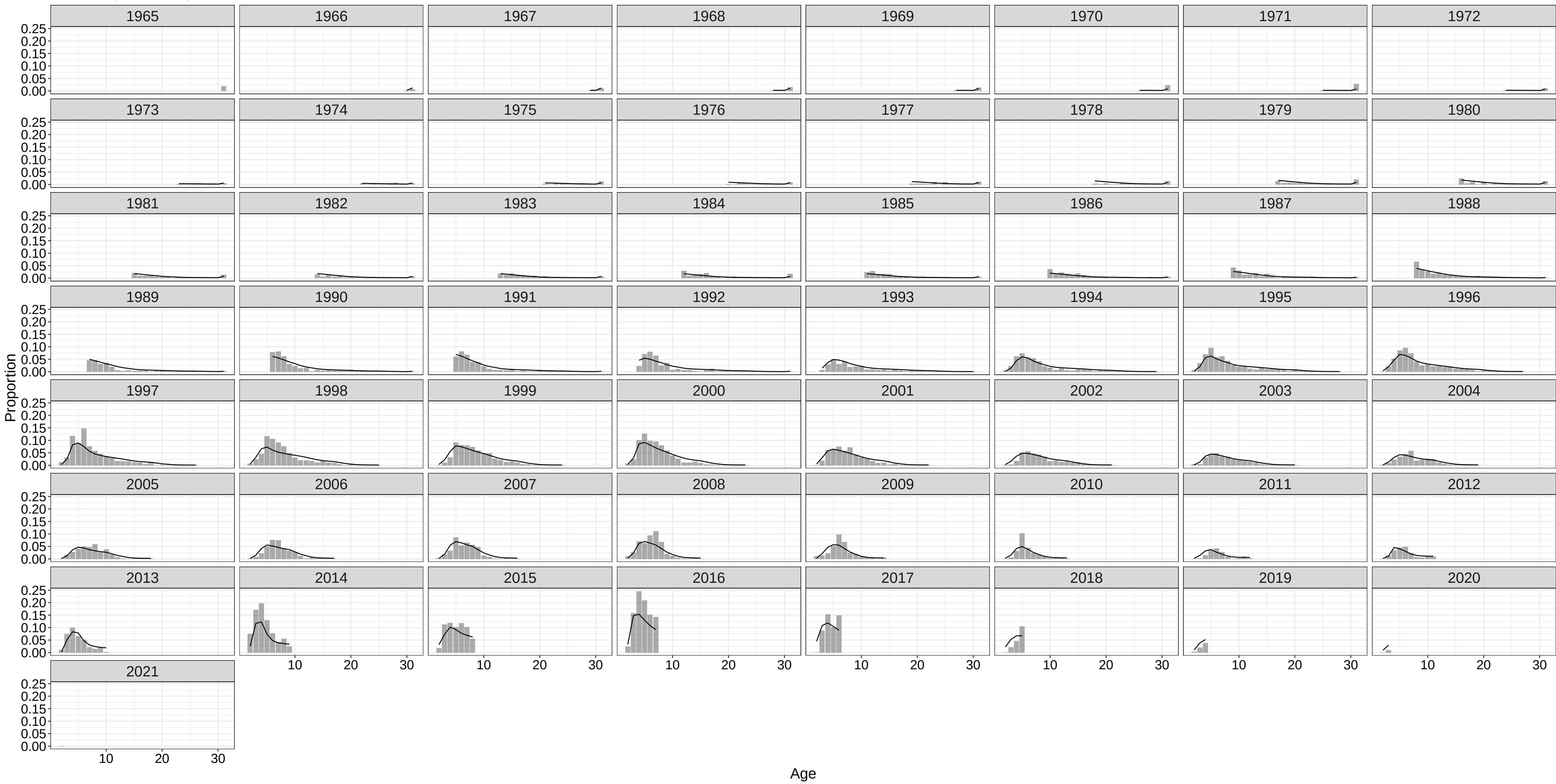
TF Length Comps Male



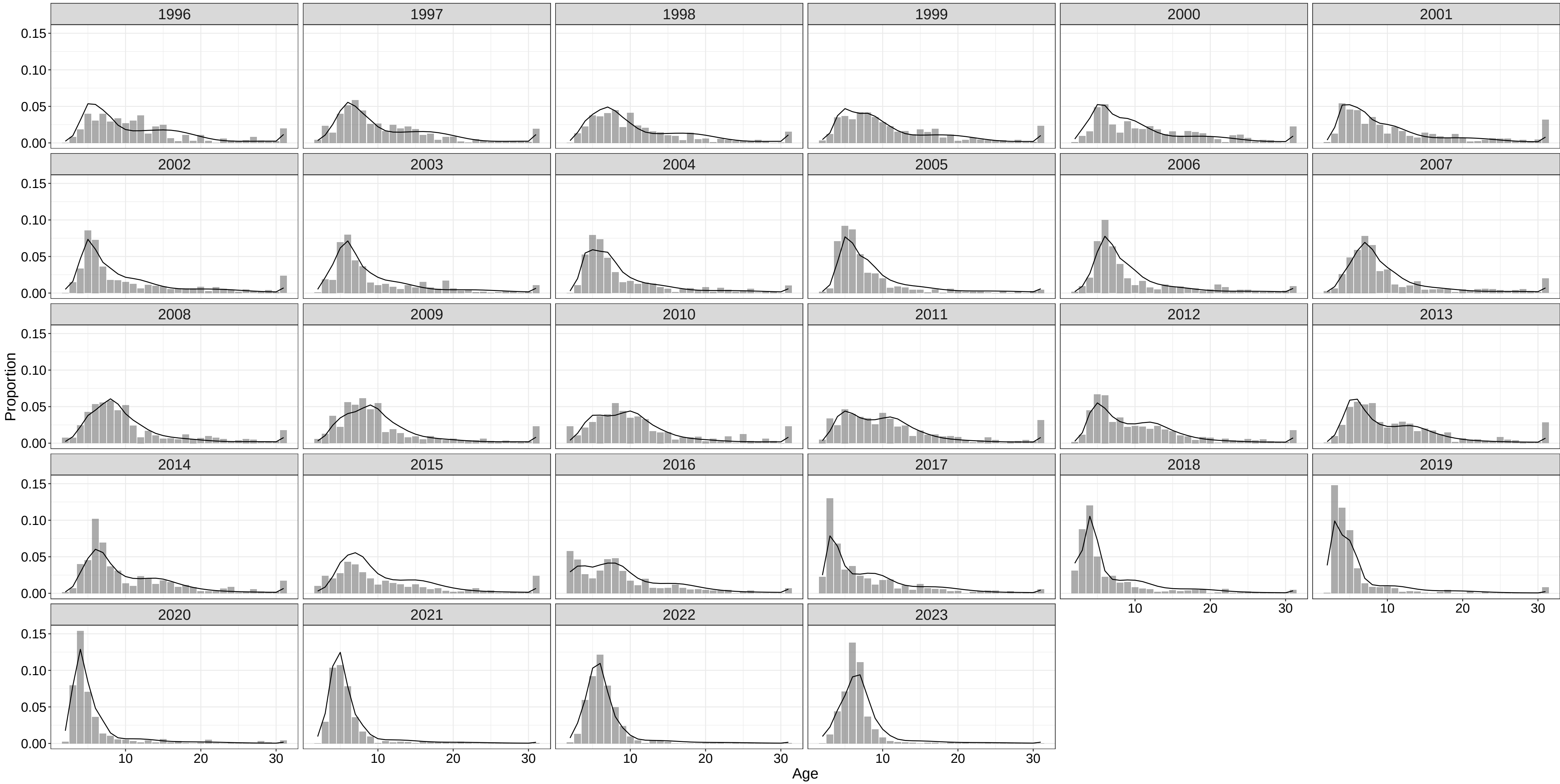
LLS Age Comps Female



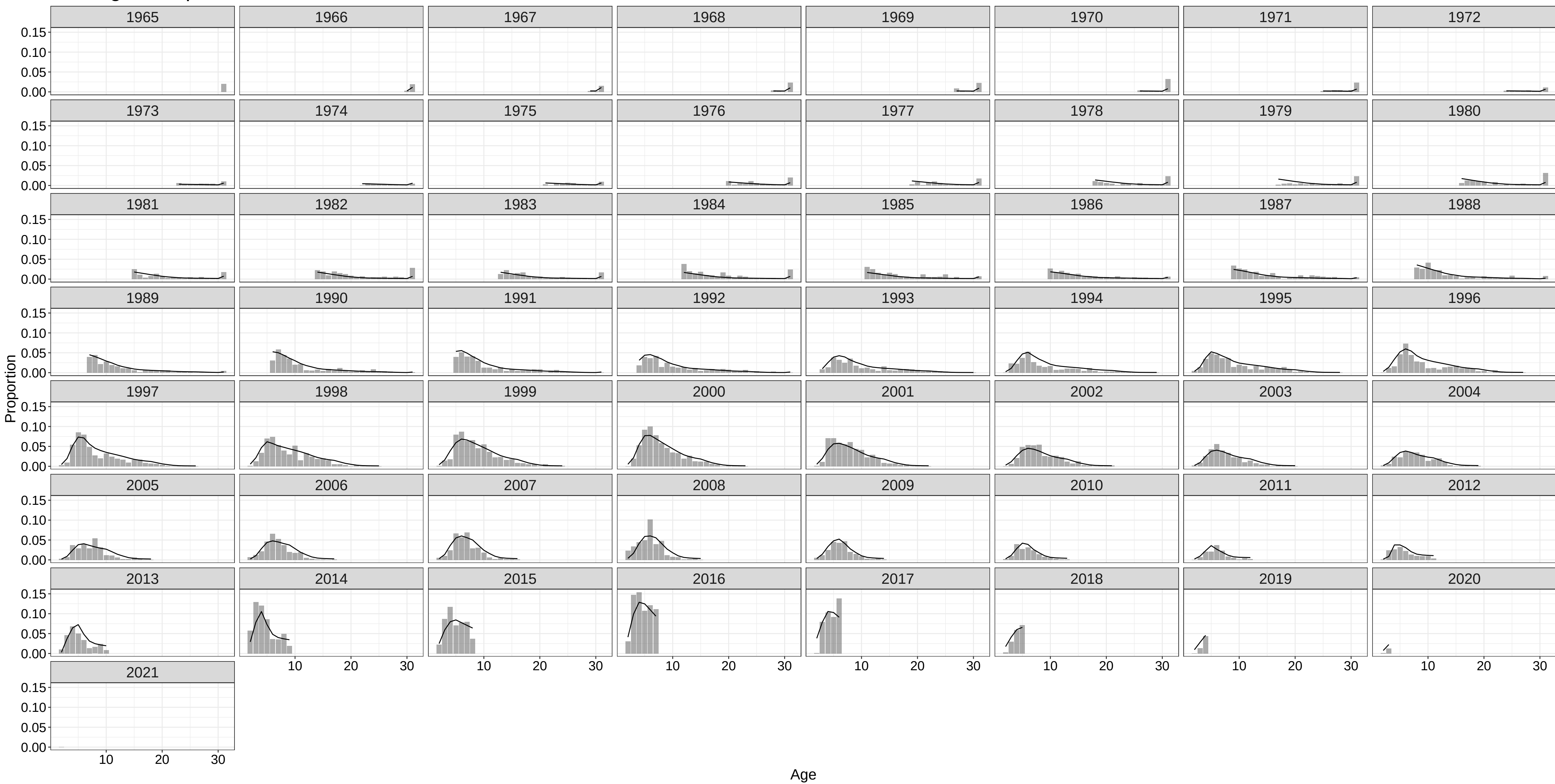
LLS Age Comps Cohort Fit Female



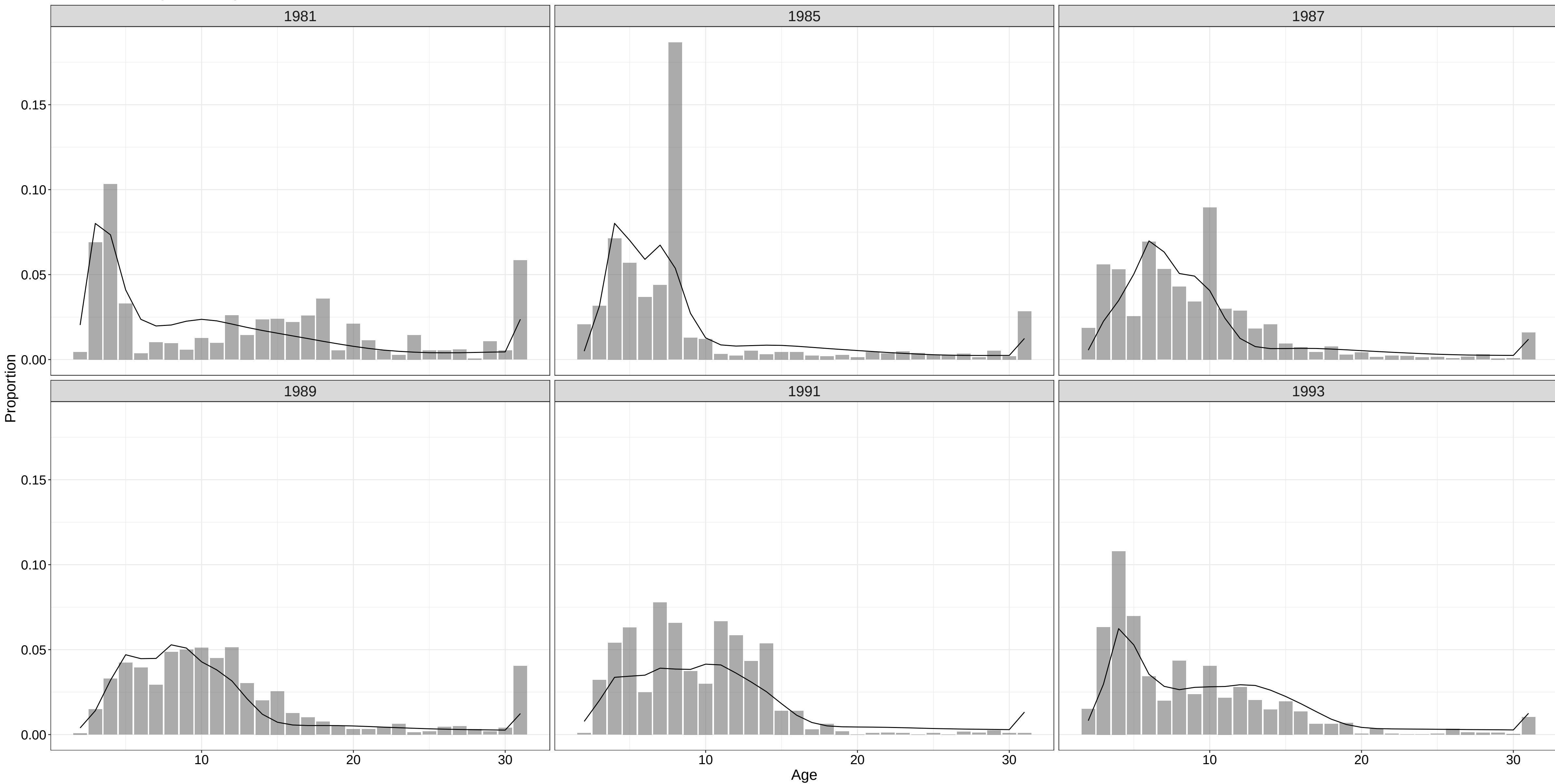
LLS Age Comps Male



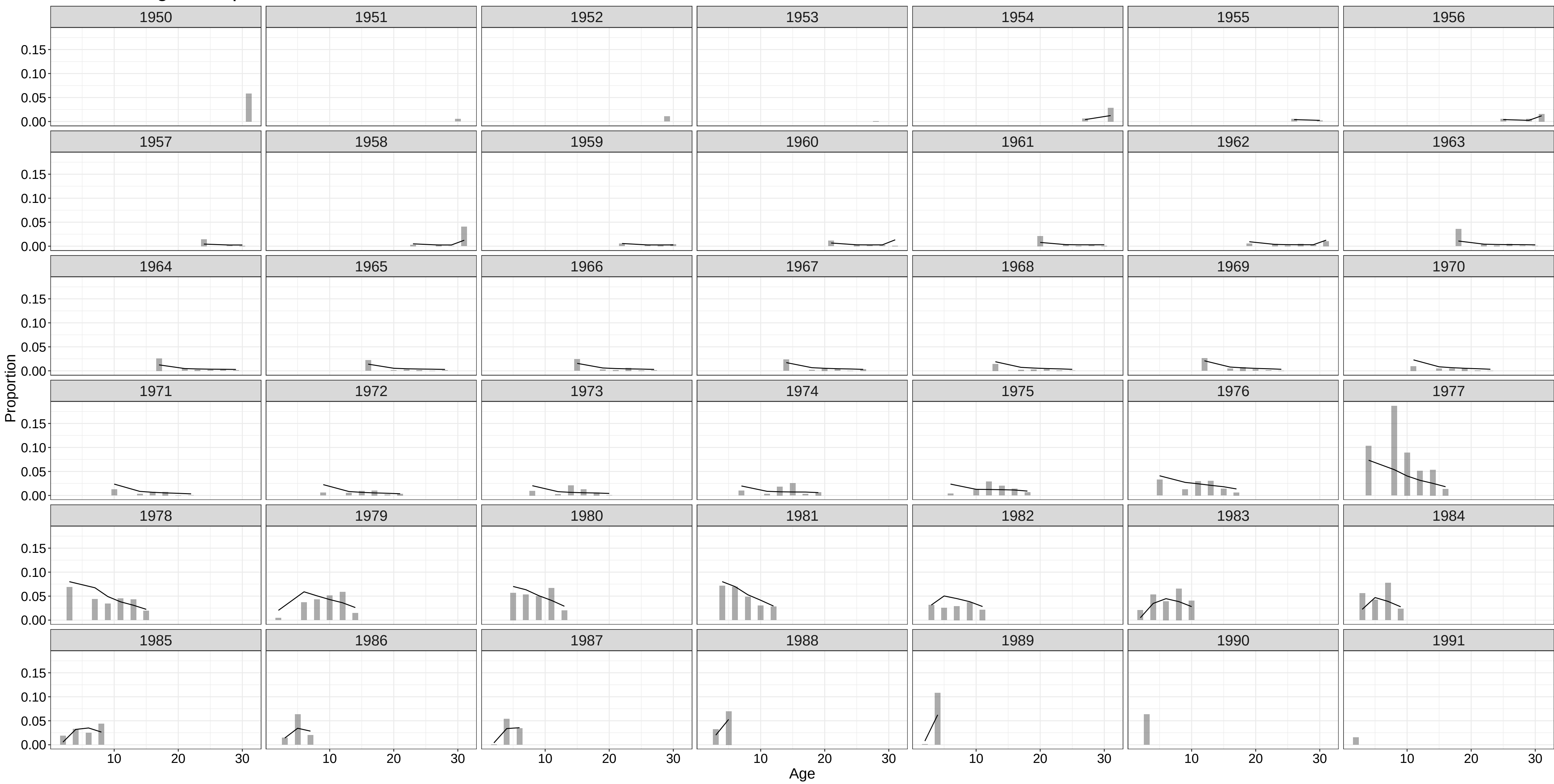
LLS Age Comps Cohort Fit Male



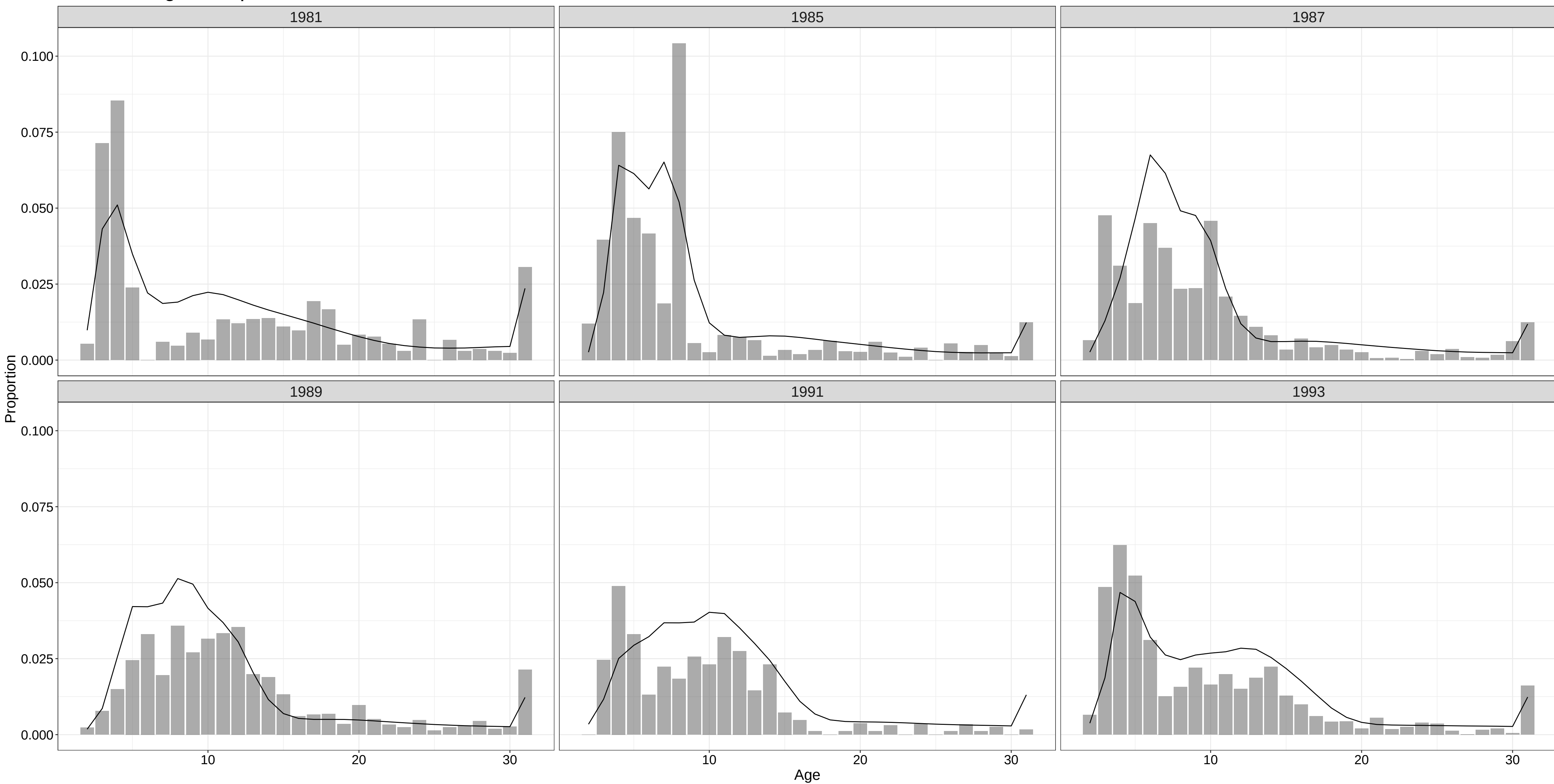
JPN LLS Age Comps Female



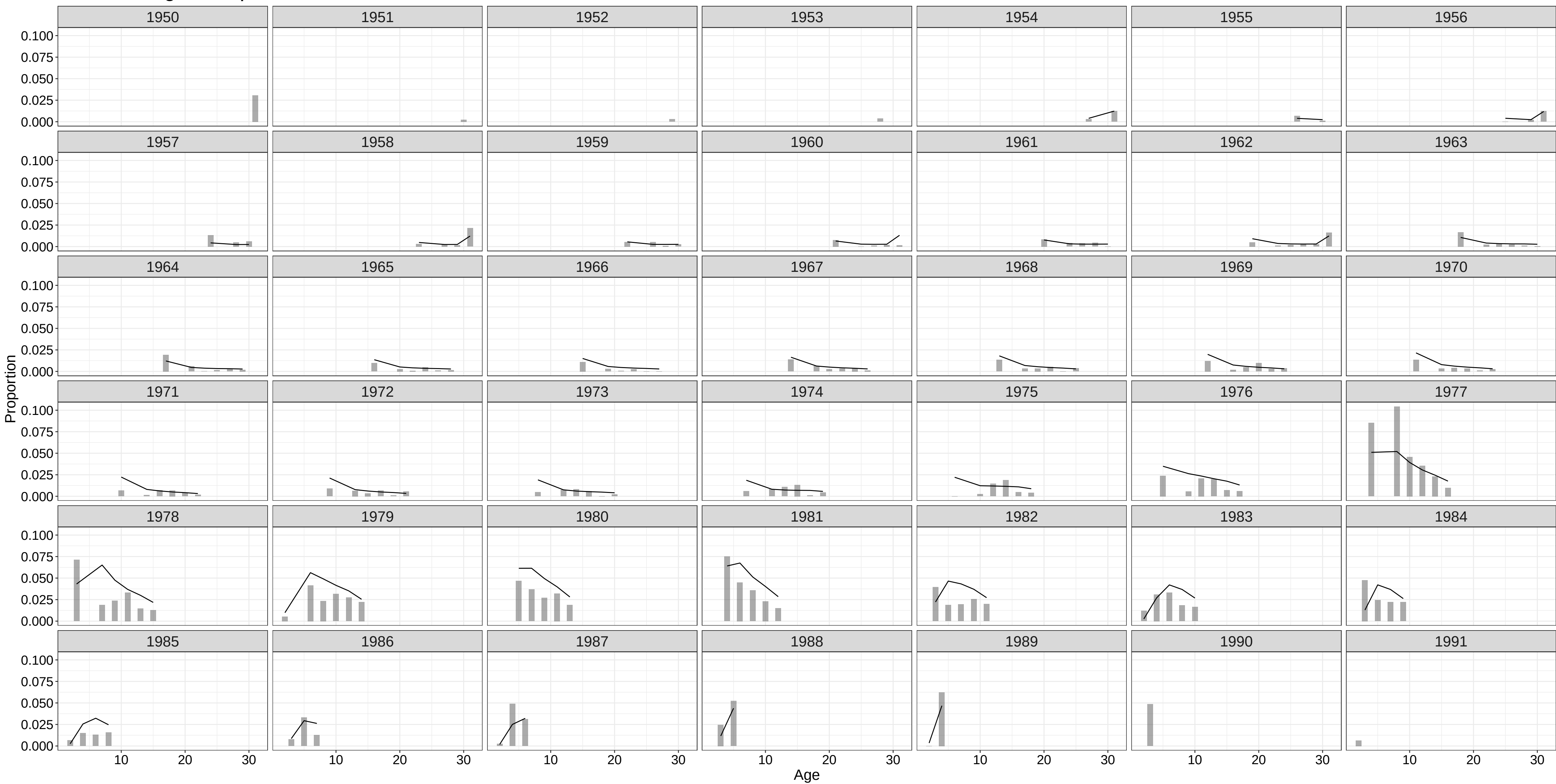
JPN LLS Age Comps Cohort Fit Female



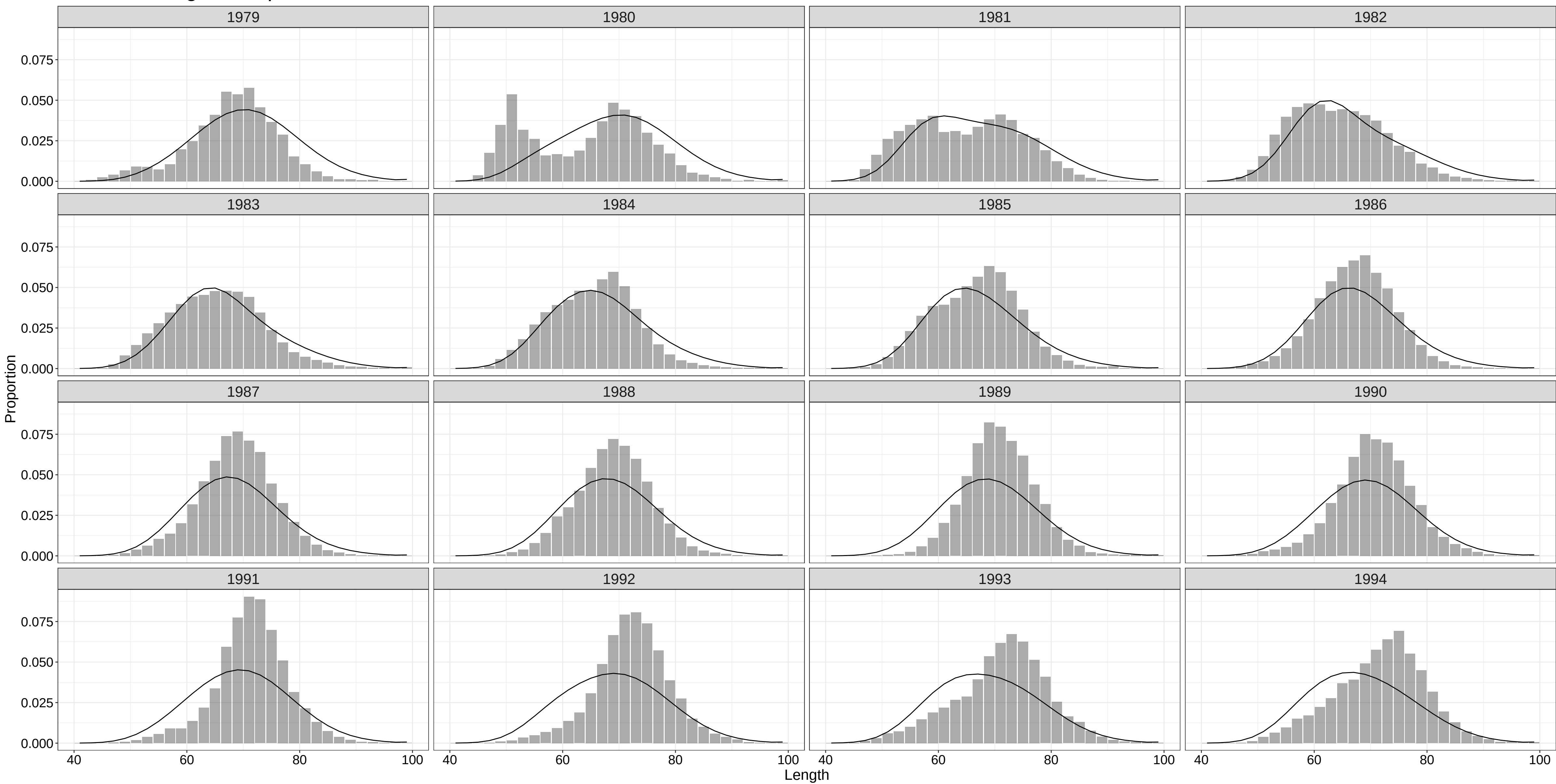
JPN LLS Age Comps Male



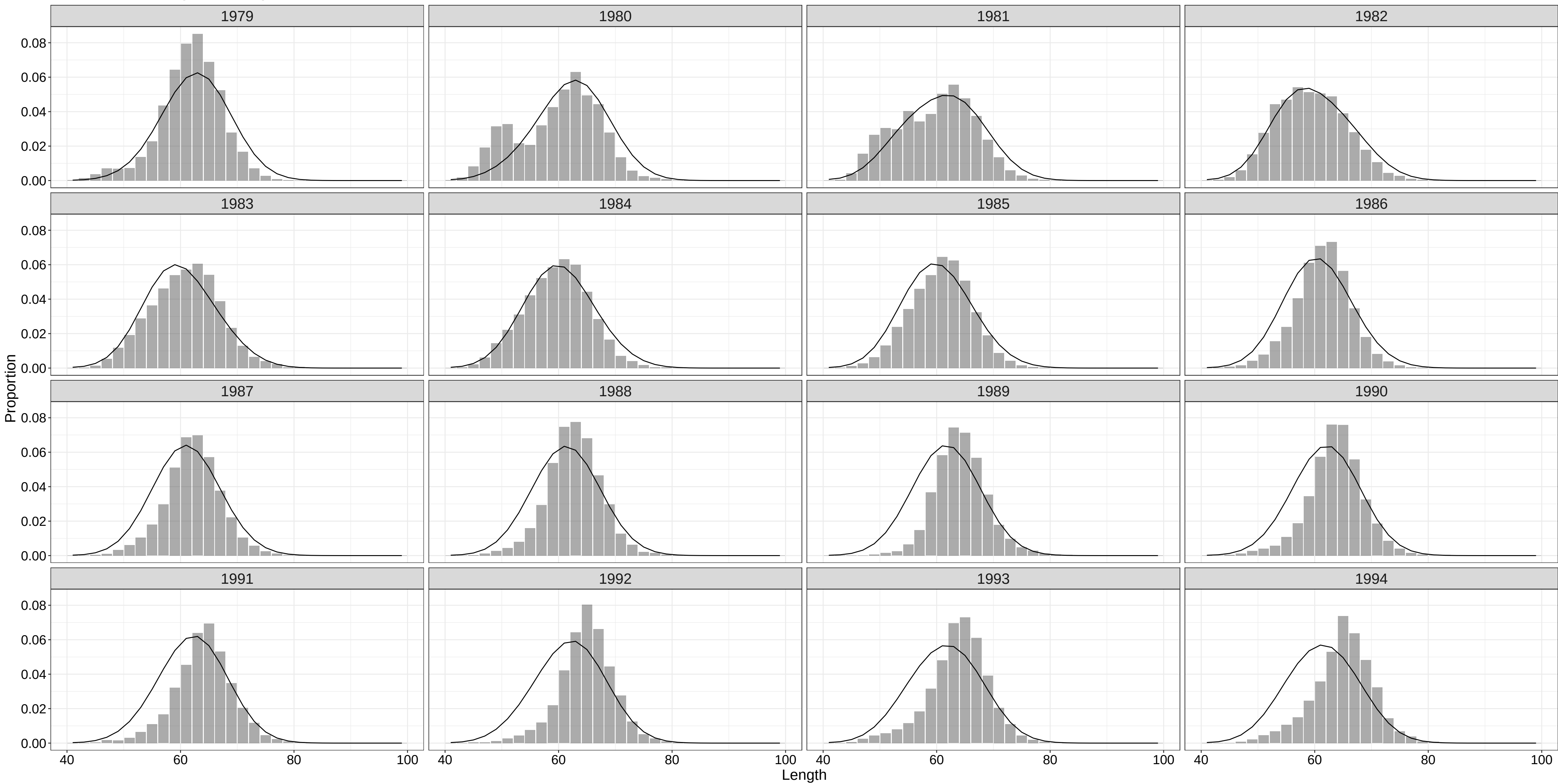
JPN LLS Age Comps Cohort Fit Male



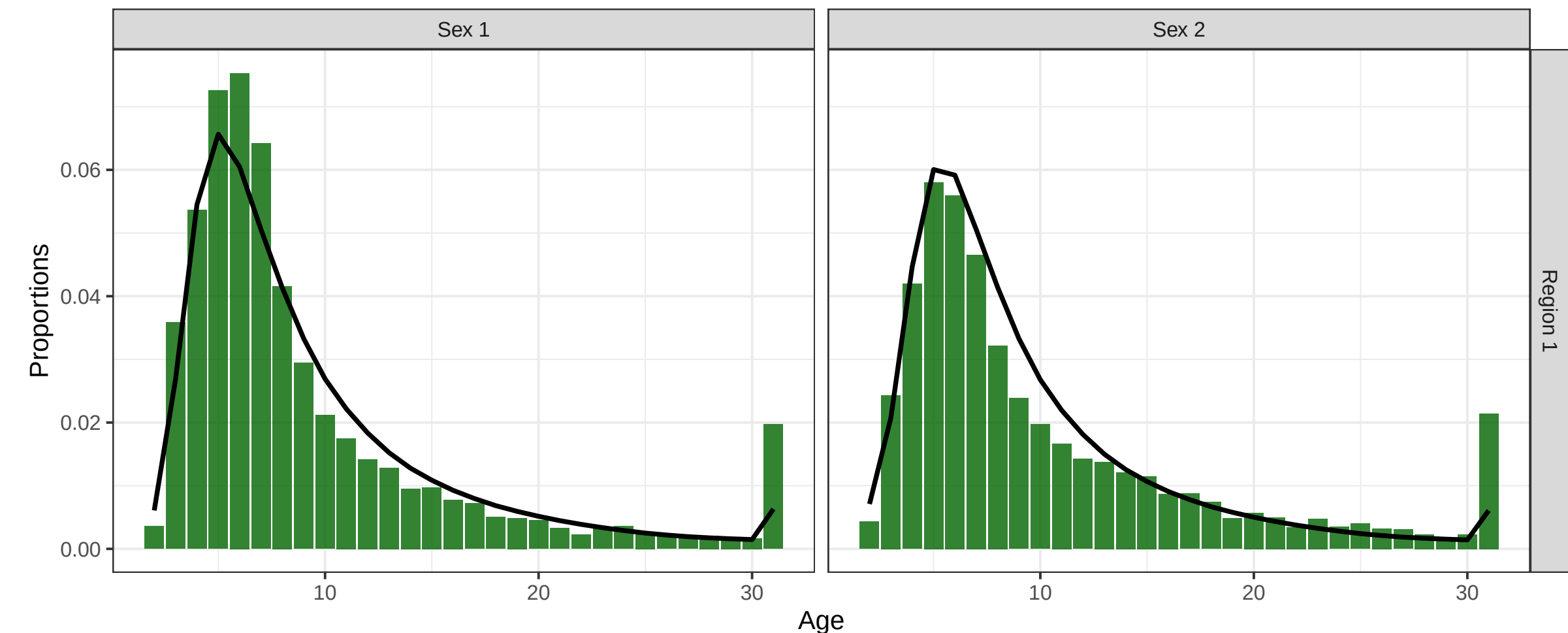
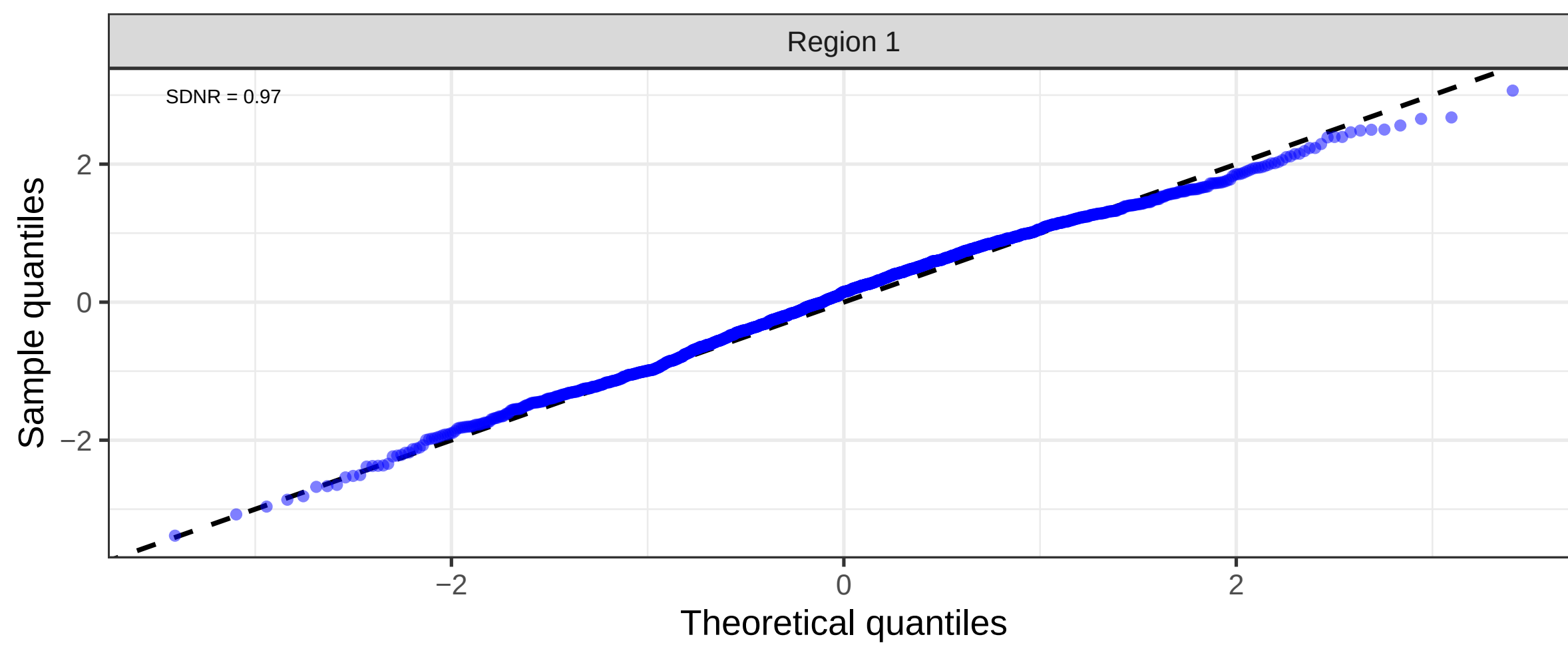
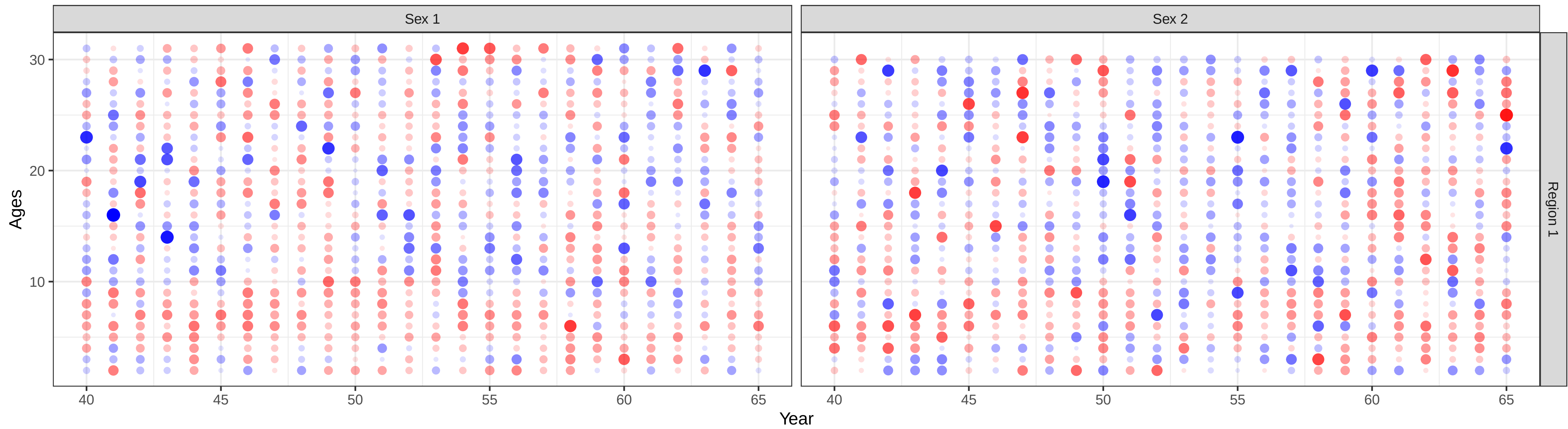
JPN LLS Length Comps Female

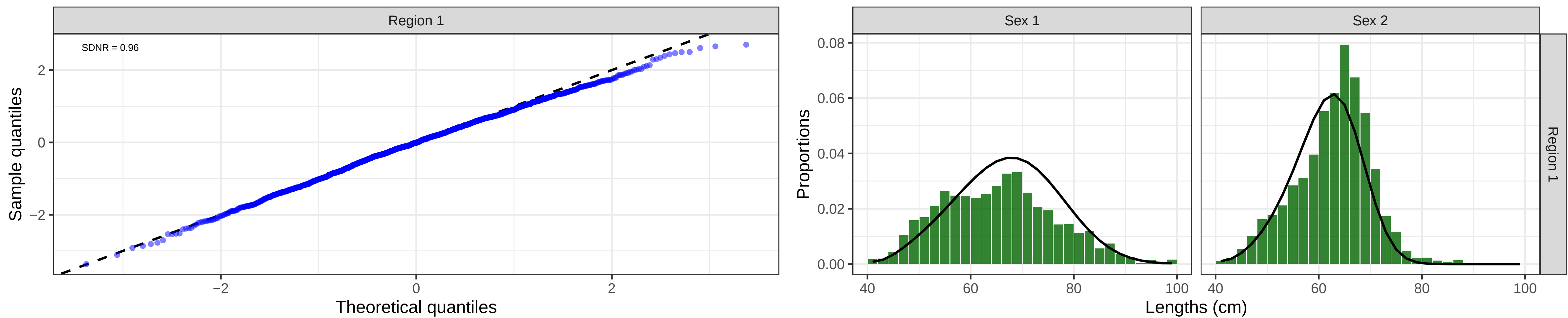
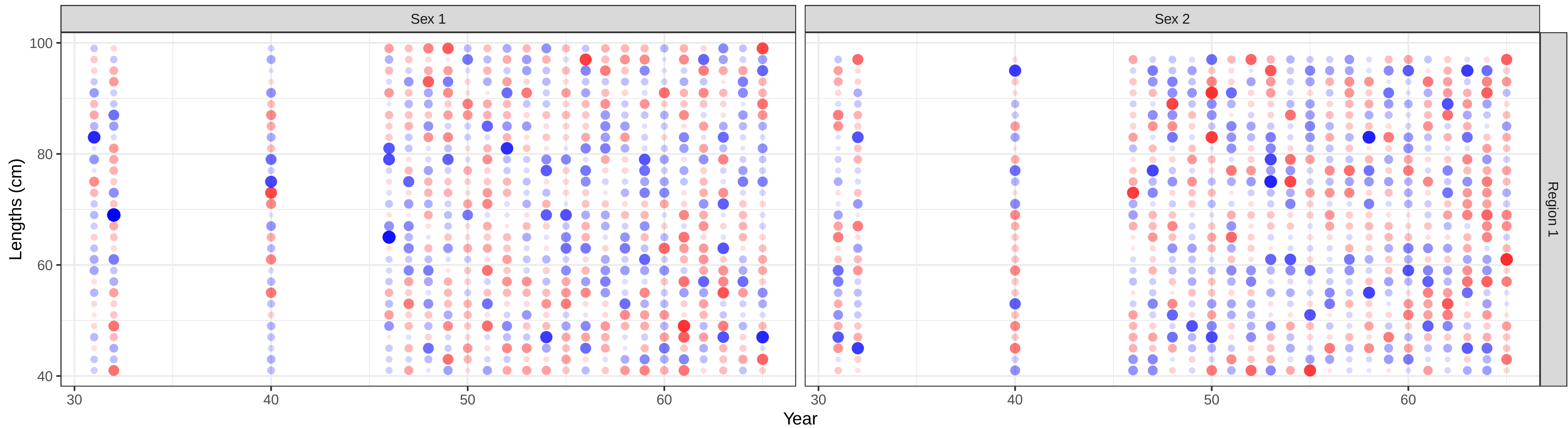


JPN LLS Length Comps Male

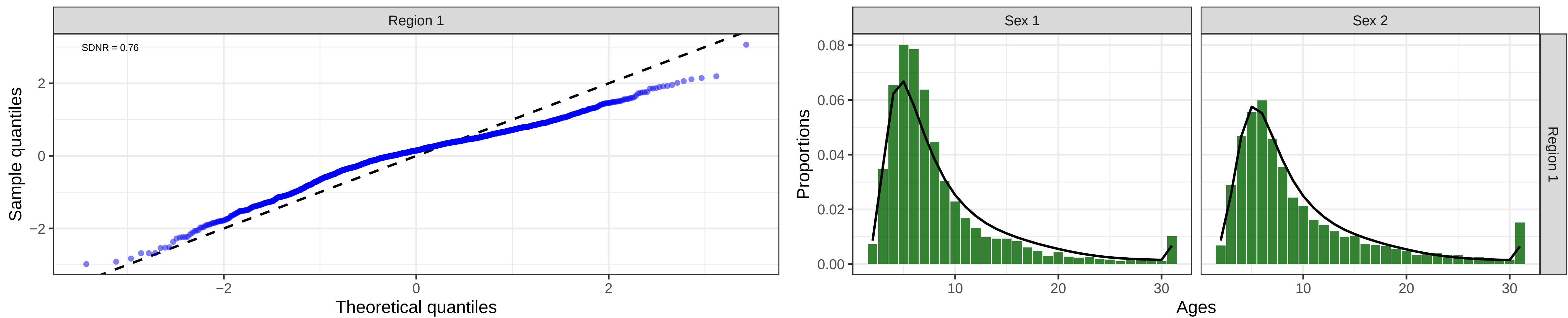
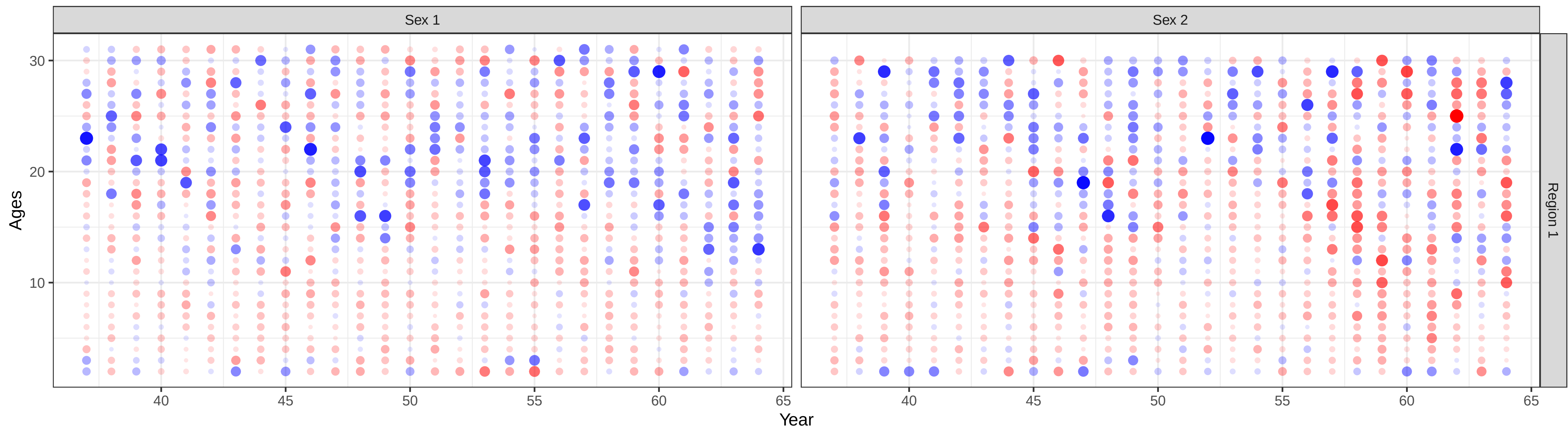


abs(Resid) ● 1 ● 2 ● 3 Sign ● Neg ● Pos

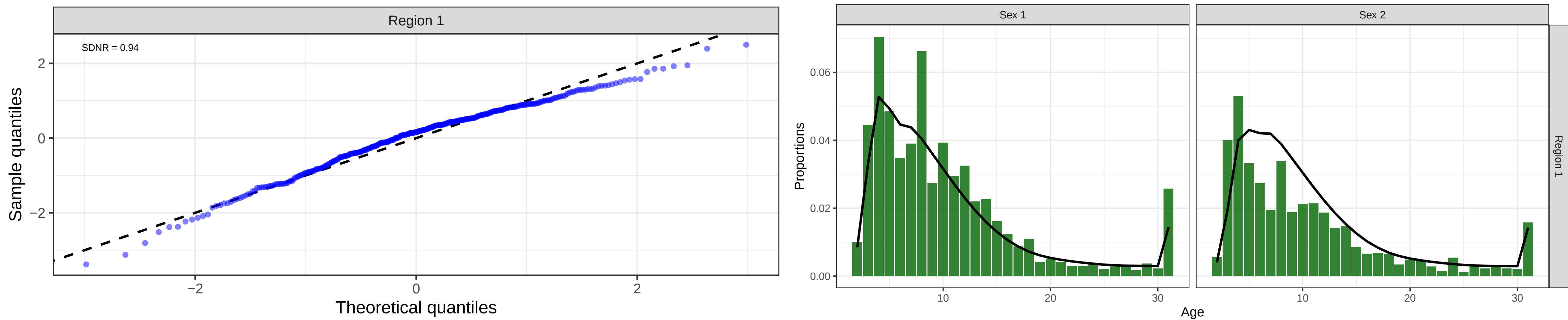
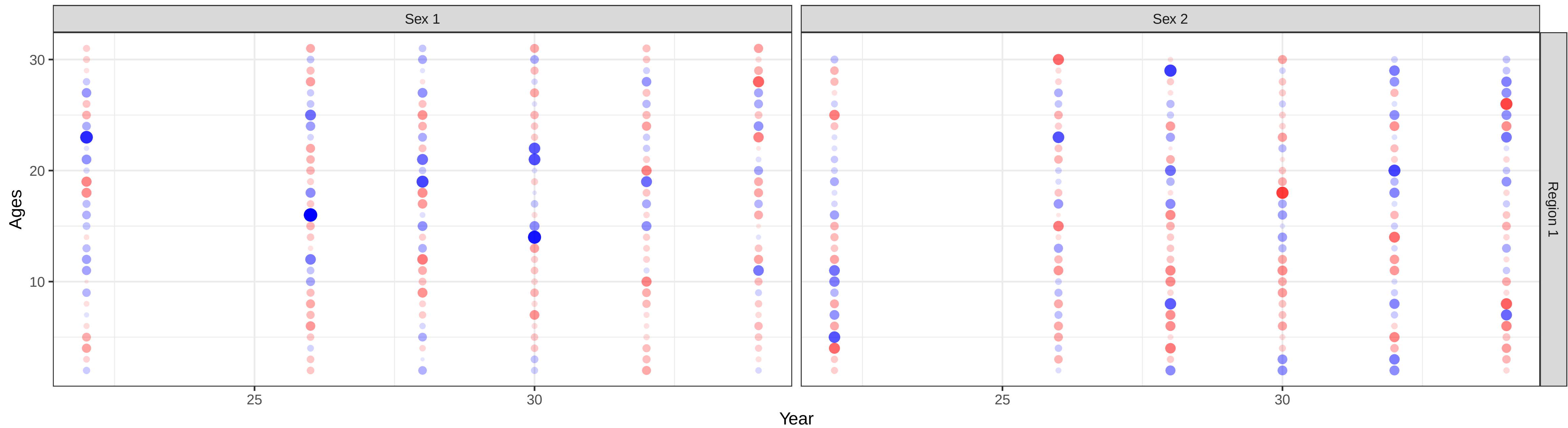




abs(Resid) 1 2 3 Sign ● Neg ● Pos



abs(Resid) 1 2 3 Sign ● Neg ● Pos



Sign ● Neg ● Pos abs(Resid) ● 1 ● 2

